

A CNN–RF hybrid approach for deforestation detection using Google Earth Engine satellite imagery

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ABSTRACT

This paper presents a lightweight and hybrid deep learning strategy that combines Convolutional Neural Networks (CNNs) and Random Forest (RF) classifiers for Land Use and Land Cover (LULC) mapping and detecting changes in the district of Bengaluru, Karnataka, India, between 2018 and 2024. The proposed model utilizes Google Earth Engine (GEE) to operate on multi-spectral Sentinel-2, Sentinel-1 radar backscatter, and topographic features such as elevation and slope. More specifically, the developed deep learning strategy applies the CNN module to efficiently capture the spatial and spectral details, while the RF classifier enhances decision-level accuracy through ensemble learning. Our findings report an accuracy of 98.48% and a Kappa coefficient of 98.10% for the year 2018 and an accuracy of 99.26% and Kappa coefficient of 99.08% for the year 2024, and outperform individual CNN, RF, and traditional ensemble models. Therefore, the proposed model presents a feasible and scalable solution for land use planning and forest monitoring in variable landscapes.

1. Introduction

One of the main environmental concerns in the world today is deforestation. Deforestation has been identified as one of the main environmental challenges in the rapidly urbanizing areas of developing countries. The rate of deforestation has been observed to be highly correlated with the expansion of infrastructure in these areas (Foley et al., 2005). The detection of deforestation has been considered critical in the sustainable management of the environment. The use of satellite remote sensing has been identified as the main solution in the detection of deforestation. The use of multi-source data from the Sentinel-1 Synthetic Aperture Radar (SAR) and the Sentinel-2 multispectral images has been considered useful in the provision of spatial, spectral, and structural information for the classification of LULC (Gong et al., 2013).

With the latest advancements in deep learning techniques, such as Convolutional Neural Networks (CNNs), the accuracy of LULC classification has been improved significantly (Yang et al., 2019). However, the majority of the deep learning algorithms are

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computationally expensive and require a large amount of training data and high-end hardware, which makes them difficult to apply in resource-constrained environments (Ma et al., 2019). Classical machine learning algorithms such as Random Forest (RF) and Support Vector Machine (SVM) are computationally efficient but unable to effectively identify the complex spatial and spectral information necessary for accurate deforestation detection. Although the application of a hybrid CNN-RF model has been proposed, the majority of the existing methods rely on deep learning models with fully connected layers, which makes the algorithms computationally expensive (Wang et al., 2023).

To mitigate these issues, this research aims to introduce a lightweight CNN-RF hybrid model to efficiently and effectively detect deforestation with multi-source satellite images (Winkler et al., 2021). In this research, the proposed model is based on a lightweight two-layer CNN architecture without fully connected layers. This approach is implemented to reduce the number of trainable parameters while still allowing the model to learn important features (Gottlieb and Mankin, 2024). These features are then classified by an RF classifier to achieve a good balance between high classification accuracy and computational efficiency. The objectives of the study include the following: (i) the development of an efficient lightweight CNN-RF hybrid model, (ii) the integration of multiple sources of satellite data for feature representation, (iii) the extraction of spatial-spectral context using patch-based learning, (iv) the achievement of high classification accuracy with reduced model complexity.

This model is suitable for deployment on resource-constrained devices. The integration of multiple sources of data is done using Sentinel-1 SAR data, Sentinel-2 optical data, and ALOS World 3D-30 m (AW3D30) topographic data (Zhao et al., 2023a). The robustness of the classification is improved using the patch-based approach with the dimensions (15x15x11). The proposed methodology is applied to the Bengaluru district in the Indian state of Karnataka using data from the years 2018 and 2024. The changes in the land cover over time between the five LULC categories of Forest, Plantation, Fallow Land, Water Body, and Built-up Area are studied (Babitha et al., 2025a).

The key contributions of the present work are: (1) the development of a lightweight CNN-RF hybrid architecture without fully connected layers; (2) the fusion of multiple source satellite data; (3) the validation of the proposed architecture over time with its overall supremacy over different state-of-the-art models such as CNN, ANN, SVM, RF, Traditional CNN-RF, MobileNetV2+RF, and XGBoost.

This section clearly indicates the research gap in the existing body of literature on detecting deforestation. The research gap in this context is the lack of lightweight CNN-RF hybrid models that can be deployed at the edge and process multi-source satellite data in the GEE platform. The contributions of the current study are: lightweight CNN-RF hybrid models without the need for fully connected layers to minimize computational overhead; the fusion of multi-source data such as Sentinel-1, Sentinel-2, and AW3D30 in the GEE platform; and the validation of the proposed approach on two temporal datasets: 2018 and 2024 for the Bengaluru region.

The scope of this study is restricted to one urban fringe study area in Karnataka, India, and two temporal snapshots. The generalizability of the proposed framework in other geographic areas, biomes, and/or satellite sensor configurations needs to be verified in future studies.

The rest of the paper is structured as follows: Section 2 discusses the background; Section 3 discusses the methodology; Section 4 discusses the results and their discussion; Section 5 discusses the limitations and the scope of the work; and Section 6 concludes the paper.

2. Related works

Deforestation detection has assumed importance in India with rising anthropogenic pressures and rapid urbanization, especially in ecologically fragile areas. Several studies have demonstrated the application of remote sensing methods in detecting forest cover transitions and tracking forest degradation, utilizing Moderate Resolution Imaging Spectroradiometer (MODIS) time-series data to identify deforestation hotspots in central India (Bhuyan et al., 2025; Das et al., 2025; Papoutsis et al., 2023). These studies have also highlighted the importance of temporal frequency in detecting forest disturbances. Likewise, Landsat imagery and Normalized Difference Vegetation Index (NDVI)-based classification to measure forest cover loss in Odisha (Babitha et al., 2025b), illustrating the role of vegetation indices in capturing change dynamics. Since the advent of more high-resolution datasets such as Sentinel-2 and cloud computing platforms such as GEE, research has been more regionalized and mechanized. A research work applied Sentinel-2 imagery in GEE to map forest loss in the Western Ghats with good accuracy via RF classifiers (Sathyanarayanan et al., 2020). Such methods, however, tend to rely on conventional ML algorithms, which can be devoid of the spatial contextual wisdom of complex forest environments.

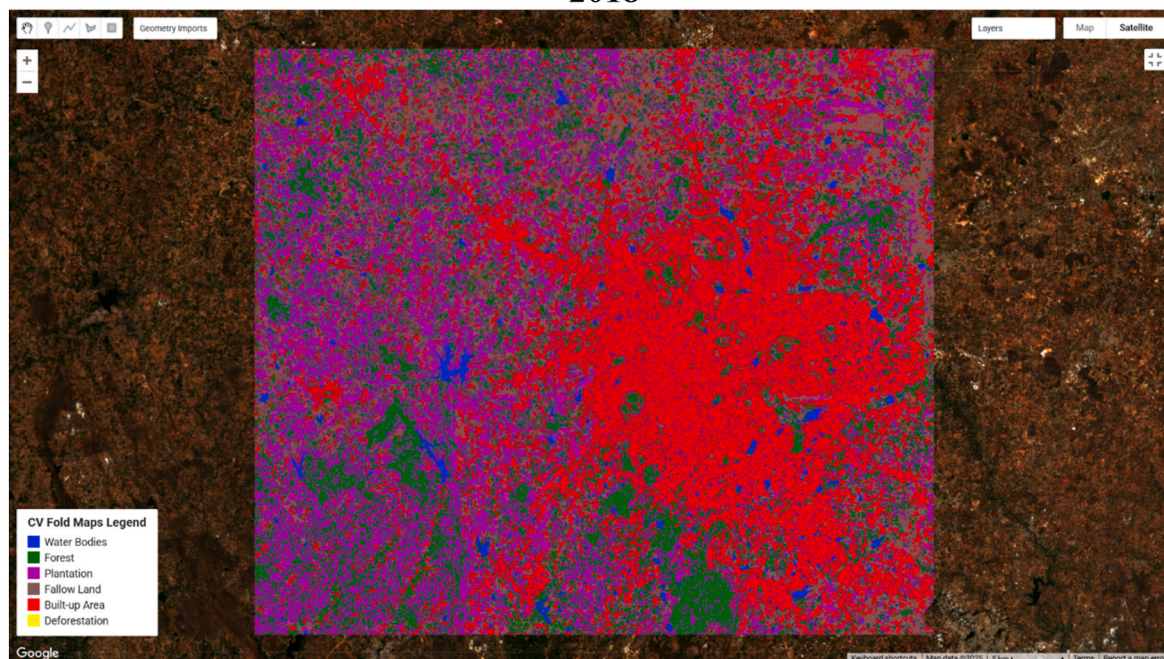
Emerging advancements have also brought deep learning models to forest monitoring. Deep learning models have also been incorporated due to the emerging advancements. In this regard, the CNN-based approach has been utilized for the detection of patches of deforestation in Jharkhand using the fused Sentinel-1 and Sentinel-2 images. The findings showed better discrimination between dense forest cover and plantation, as well as scrubland, regions (Bergamasco et al., 2023). However, most deep learning models are still computationally demanding, which hinders their deployment at scale. In spite of these limitations, few studies have covered light-weight hybrid models that can effectively identify deforestation with efficient handling of satellite data in bulk (Li et al., 2022; Ma et al., 2024; Roy et al., 2023). Additionally, research on urban periphery forests like Bengaluru and Karnataka in India is lacking. This research bridges the gap by proposing a lightweight CNN-RF hybrid model specifically designed for detecting forest change in rapidly evolving urban-forest boundaries.

Recent developments in Earth observation and computational modeling have greatly improved the monitoring of deforestation and land cover (Deng et al., 2025). In order to balance the power of CNNs in feature learning and the efficiency of traditional classifiers, various works introduced hybrid CNN-RF models that utilized a CNN-RF model to enhance classification in the presence of limited

training data. They utilized deep residual CNNs with a subsequent RF, but their model needed a lot of computation and memory, which made it infeasible for edge deployment. Parallel trends emerging within the 2020–2023 window emphasize hybrid CNN-RF architectures (Lee et al., 2020). An investigation using Sentinel-2 time-series for early crop mapping demonstrated that CNNs are superior in deriving informative spatial features, while RF classifiers are more stable with small training sample sizes, resulting in more consistent outcomes in the face of data insufficiency.

Another use in burned-area mapping combining optical and SAR data further showed CNN-RF hybrids' strength compared to

2018



2024

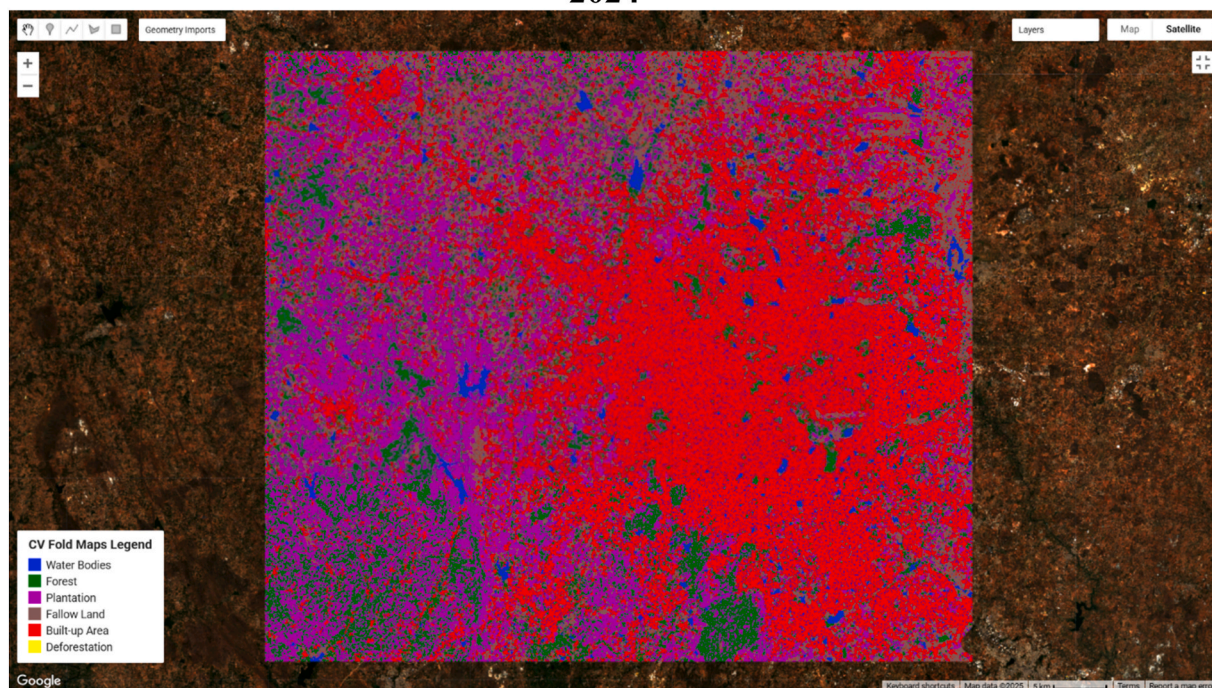


Fig. 1. Study area: - Bengaluru - Karnataka - India-for the year 2018 and 2024 Captured from: Google Earth Engine Map (Map data ©2023 Google, INEGI, Maxar Technologies).

independent CNN or RF models, where RF successfully substituted the softmax layer and enhanced prediction stability (Wang et al., 2021). The work carried out in the Annamalai Hills of Tamil Nadu utilized a hybrid ML pipeline comprising RF and other ensemble techniques for land cover change detection without utilizing deep CNN features (John and Zhang, 2022). Therefore, although both deep and traditional ensemble techniques have been effectively coupled together in agricultural and fire contexts, their integration to apply to extensive LULC classification and temporal change detection for multiple land cover types has yet to be comprehensively explored (Berger et al., 2012).

LULC classification and permanent change detection are suitably focused on the development of a permanent procedure for monitoring changes in land cover. The Continuous Change Detection (CCD) method uses a dual-phase methodology to eliminate the clouds and snow for accurate observation and appropriate assessment. With all the accessible Landsat data, frequent data acquisition can be generated to support continuous prediction of alterations in landscapes like farming alteration, forestland degradation, urban sprawl, etc. CCD also categorizes the information according to the measurement of changes based on the observed spectral behavior of the data obtained during the period. The efficiency of the approach demonstrates the capability of detecting the changes that may not be detected by the conventional methods (Drusch et al., 2012). This CCD approach is particularly helpful in frequent cloud-covering regions due to its clear observations, environmental management, and land monitoring measures.

Taking note of the limitations of deep models, recent studies have been focusing on light-weight architectures proposed a light-weight CNN for Sentinel-2 based land cover classification with evidence of decreased training time and slight performance reduction proposed SpecSAR-Former, a light-weight transformer structure that combines Sentinel-1 and Sentinel-2 data for global LULC mapping (Ienco et al., 2019). Their model ensured high accuracy and feasibility for implementation in resource-constrained environments. Also, demonstrated that ensemble RF models with combined Sentinel-1 and Sentinel-2 data could achieve high classification accuracies (96.9%) with no need for deep learning. This again establishes the merit of multisensory integration, particularly for change detection applications in urban-rural contexts like Bengaluru (Richard and Navin, 2024).

Light-weight hybrid architectures for edge and mobile applications are also investigated in recent advances and introduced Automated Land Cover (AutoLC), a remote sensing LULC task-optimized neural architecture search system, presented Ring Mo-Lite, a CNN-transformer hybrid architecture for model size reduction without compromising performance, which provides new avenues for mobile deployment in forestry and agriculture (Mathew et al., 2024).

In spite of all these developments, only a handful of studies have proposed compact CNN-RF hybrids with multi-source input (Sentinel-1, Sentinel-2, and topography) for deforestation observation in fast-growing urban centers like Bengaluru (Dehingia et al., 2022; Zhao et al., 2023b). This research bridges that gap by presenting a light-weight hybrid model with the ability to deploy on an edge and superior classification accuracy for five land cover classes. These works lay a solid groundwork, showing how light-weight CNN-RF hybrids can improve model accuracy and robustness—especially in low-data situations. This paper builds on this frontier by implementing a hybrid approach in multi-class LULC mapping (forest, urban, fallow, water, and plantation) and change quantification within the GEE framework in Karnataka.

The literature review offered in this section of the paper makes the following specific contributions to the study. It critically and systematically reveals three fundamental research gaps in existing works on deforestation detection and LULC classification, namely, the scarcity of lightweight CNN-RF hybrid architectures that are optimally designed for edge device deployment in tropical deforestation detection, the insufficient utilization of multi-source satellite imagery combining Sentinel-2 optical, Sentinel-1 SAR, and AW3D30 topographic features within a single GEE-based classification workflow, and the insufficient benchmarking of state-of-the-art lightweight deep learning architectures against traditional RS-based classifiers in multi-temporal LULC classification scenarios. The literature review establishes the theoretical and empirical basis for the methodological choices and decisions taken in this study, as it relates to the proposed approach, by critically engaging and drawing from existing literature. The identified gaps are the primary drivers of the proposed approach for a lightweight CNN-RF hybrid classifier, as detailed in the subsequent sections of this paper.

The scope of the literature review is limited in the following ways. Firstly, the review is primarily focused on supervised patch-based classification and hybrid deep learning models, while there is limited exploration of the full scope of unsupervised, semi-supervised, and self-supervised approaches to change detection, which is considered to be emerging literature. Secondly, the variety of geographic settings, scales, sensor types, and accuracy evaluation protocols in the literature reviewed makes it difficult to compare the accuracy results quantitatively.

3. Methodology

3.1. Study area

The area around the city of Bengaluru is taken as the study area, which is the capital of Karnataka, India, is shown in Fig. 1 Bengaluru is a fast-growing metropolitan area under severe urbanization and consequent land use alteration, particularly deforestation in its nearby forest edges. The study location falls between latitudes 12.7°N and 13.2°N and longitudes 77.3°E to 77.8°E and covers urban areas, peri-urban forests, agricultural fields, and reserve forest patches. The area offers an active terrain on which to evaluate temporal land cover changes between 2018 and 2024 through satellite-based classification models.

3.2. Data acquisition

A worldwide Digital Surface Model (DSM) dataset named Advanced Land Observing Satellite AW3D30 was employed herein (Gorelick et al., 2017). This dataset has a horizontal resolution of about 30 m and is built on the World 3D Topographic Data's DSM

dataset (5-m mesh version). The update released in January 2021, identified as Version 3.2, significantly improved the data by re-estimating high altitude fields, combining additional improved information, and processing with a more sophisticated processing procedure. Preferably, the update integrated an implementation that utilized alternative pixel spacing between different zones of latitudes in the high-altitude field (Babitha et al., 2023b). The additional coastline data dataset was updated, and fresh auxiliary data were included. Furthermore, the procedure for flagging unique values in the data was considerably improved. Here, the AW3D DSM elevation is computed by an image-mapping algorithm that utilizes a stereo pair of optically generated images (Ortega et al., 2021; Bouvet et al., 2018). In this processing, clouds, snow, and ice are recognized automatically and masked. The possibility remains that some mismatched points may occur, primarily around or on the border of these features. Such mismatched points have the potential to create elevation errors in the resultant DSM product.

In this study, satellite data from Sentinel-1 (S1), Sentinel-2 (S2), and slope and elevation were obtained and preprocessed to aid robust LULC classification and change detection in Karnataka. The model utilizes hyperspectral, radar, and topographic data sets, which provide a complete picture of surface dynamics over the years 2018 and 2024. Data were retrieved through the GEE platform, where there is cloud-based access to time-series satellite data with integrated pre-processing tools. The GEE offers a cloud-based platform with access to large-scale geospatial data sets and parallel computing capabilities (Benedetti et al., 2018). This enables the efficient handling of extensive satellite imagery data without the need for any local computational resources. In the proposed study, the GEE is used for scalable data acquisition and preprocessing, which improves the efficiency of the proposed CNN-RF model. In order to maintain data quality, a stringent cloud cover requirement of fewer than 10% clouds was used for every year of imagery (Xu, 2008).

The following data sets were employed:

Sentinel-1 (SAR): Offers structural and backscatter data useful for the identification of built-up areas and water bodies. Sentinel-1 offers dual-polarized SAR data (VV and VH bands) in C-band, not influenced by clouds or atmospheric conditions. With a revisit period of 6–12 days and a 10-m spatial resolution, Sentinel-1 is most effective for extracting built-up features, water boundaries, and the structure of vegetation, particularly under cloudy or monsoon conditions. The radar backscatter values were pre-processed, speckle-filtered, and then combined with optical data for classification.

Sentinel-2 (Optical): Offers multi-spectral imagery with high spatial and temporal resolution, observing vegetation, soil, and water characteristics. Sentinel-2 provides high-resolution optical imagery with 13 spectral bands of visible (RGB), near-infrared (NIR), and shortwave infrared (SWIR). One should note that important bands, such as B2 (Blue), B3 (Green), B4 (Red), and B8 (NIR), were employed for primary classification, while other bands (20–60 m) contributed to the derivation of vegetation and water indices. The data was pre-processed with the cloud and shadow masking algorithms of GEE, and seasonal median composites were created to filter out noise and temporal outliers. The primary input was Sentinel-2 ALOS AW3D30 for separating vegetation (plantation, fallow), water bodies, and built-up land.

Topographic Elevation Data: For the inclusion of terrain context into the classification process, elevation data from the ALOS AW3D30 were utilized. The global DSM of ALOS AW3D30 uses 30-m resolution elevation values from Panchromatic Remote-sensing Instrument for Stereo Mapping (PRISM) imagery. While temporally static, ALOS AW3D30 provides useful information on landform variation, which facilitates differentiation of elevation-sensitive classes like upland forest and lowland plantation (Feyisa et al., 2014). The elevation layer was normalized and resampled to have the same spatial resolution as the other inputs.

All data sets were registered, clipped to the Karnataka study region, and resampled to a uniform resolution of 10 m (where possible). The fused stack of Sentinel-1 SAR, Sentinel-2 multi-spectral bands, and ALOS elevation data served as the input tensor for the CNN model. The multi-sensor fusion helped augment the model to better differentiate between complex and spectrally confusing classes, thus enhancing classification accuracy as well as change detection consistency.

3.3. Data labelling

For the purpose of this study, a total of 2500 labelled sample points were collected across the study area, representing five different classes of LULC, i.e., forest, plantation, fallow land, water body, and built-up area, with 500 samples per class. These samples were obtained using the visual interpretation method of satellite images.

For each sample point, the spatial patch of size 15 x 15 is extracted from the multi-source satellite images, which include the features extracted from the Sentinel-1 SAR images and the topographical features. This means that the input sample is represented as a tensor of size 15 x 15 x 11, with the first two dimensions representing the spatial information and the last dimension representing the input features.

The patch-based sampling method is used to enable the CNN model to incorporate the contextual information into the samples, which can be helpful in distinguishing the different classes of LULC. Although the number of sample points is fixed at 2500, the contextual information is increased by using the spatial patch approach, which is beneficial for the CNN model in distinguishing the different classes. The dataset was divided into training sets and test sets, with 70% being used for training and 30% for testing, giving 1750 samples in the training set and 750 in the test set. This is to ensure that the model is tested using different sets, thereby avoiding the problem of overfitting.

Five major land cover classes were established for the research. The five land cover classes were chosen based on their dominance in the region, their relation to the deforestation and urbanization processes, and their separability in the multisensory data. The selection of these land cover classes strikes a good balance between the level of detail in the classification and the efficiency of the model.

- Forest: Dense and naturally growing tree cover.
- Plantation: Agricultural or cultivated tree crops.

- Fallow Land: Temporarily idle or thinly vegetated agricultural land.
- Built-up Area: Settlement, infrastructure, and hard surfaces.
- Water Body: Rivers, reservoirs, lakes, and other permanent or temporary water bodies.

To maintain temporal coherence, all the labelled data were chosen from images acquired within similar times of the year, 1st January to 31st December of the year 2018 and also 1st January to 31st December of the year 2024, to avoid phenological differences.

Spectral indices were calculated to increase the separability of land cover classes and offer further input channels to the CNN. These indices reflect vegetation vigor, soil exposure, and surface water reflectance patterns that may be less evident in single spectral bands.

NDVI: Here, the NIR light (B8) along with the Red light (B4) bands are applied, and the NDVI value 0.56 is used (Sathyanarayanan et al., 2020). Equation (1) is used for NDVI calculation.

$$\frac{B8 - B4}{B8 + B4} = NDVI \quad (1)$$

Modified Normalized Difference Water Index (MNDWI): This index is used to make water features more visible in satellite imagery without causing confusion with urban areas. It suppresses the urban surface spectral reflectance that would normally coincide with water-related features (Mcfeeters, 1996). The index is based on the green (B3), SWIR and B11 bands, which have water features tending to have strong positive index values (approximately 0.6) and urban areas having negative responses (about -0.14) (Ji et al., 2011; Lilhore, 2023). Equation (2) is used for the index calculation.

$$\frac{GREEN - SWIR}{GREEN + SWIR} = MNDWI \quad (2)$$

Normalized Difference Water Index (NDWI): To detect and emphasize water bodies in satellite images and to differentiate clearly from the neighbouring land cover categories, NDWI is used. It makes use of the green (B3) and near-infrared (NIR, B8) spectral bands, where a threshold value of about 0.38 was empirically derived based on the frequency distribution of NDWI values of water and non-water sample points distributed across the region using GEE which proves quite useful for defining the presence of surface water features (Kukreja and Kumar, 2021; Strand, 2018) and is calculated using Equation (3).

$$\frac{GREEN - NIR}{GREEN + NIR} = NDWI \quad (3)$$

Enhanced Vegetation Index (EVI): It provides improved sensitivity in regions with dense vegetation cover. It incorporates the B8, B4, and B2 spectral bands, with a spatially averaged EVI value of 0.595 for the vegetated reference pixels and is consistent with dense tropical forest canopy conditions, thereby confirming the index's sensitivity to high biomass cover (Silva et al., 2022) and Equation (4) is used for the calculation.

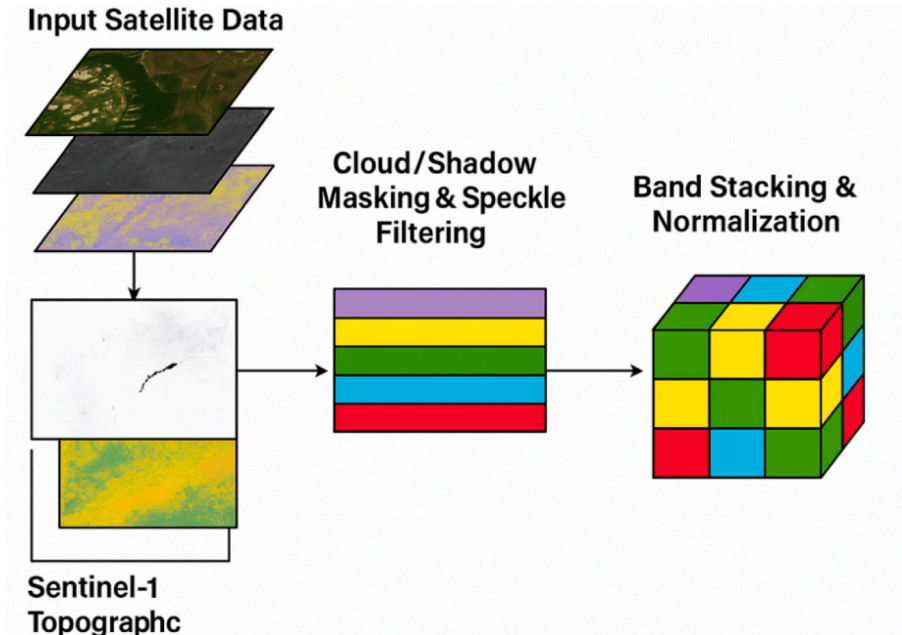


Fig. 2. Illustration of Pre-processing pipeline.

$$\frac{NIR - RED}{NIR + 6XRED - 7.5XBLUE + 1} \times 2.5 = EVI \quad (4)$$

Bare Soil Index (BSI): To improve the detection of bare soil areas, BSI is used. It was developed to detect and map regions with sparse or no vegetation cover. In this research, the index was obtained by merging the blue (B2), red (B4), near-infrared (B8), and shortwave infrared (B11) bands and is computed using Equation (5).

$$\frac{(SWIR + RED) - (NIR + BLUE)}{(SWIR + RED) + (NIR + BLUE)} = BSI \quad (5)$$

3.4. Pre-processing

While data collection is a crucial step in the analysis of images and feature extraction, overall efficacy greatly relies on using the right image pre-processing methods (Doblas et al., 2020). Pre-processing normally entails mathematical correction and normalization procedures that improve the quality of data and are integral parts of most feature extraction processes. Multi-source topographic and satellite datasets in this research were utilized to offer complementary terrain, spectral, and spatial information. Seasonal median composites from cloud-free images were produced to ensure temporal coherence and reduce atmospheric artifacts. This helped identify reliable long-term changes in land cover. Fig. 2 gives an overview of the whole pre-processing pipeline.

Sentinel-1 radar imagery (10 m resolution) was used due to its cloud-penetrating capability and sensitivity to surface roughness and wetness, employing VV, RVI, and VH polarization backscatter intensities.

Sentinel-2 optical image (10 m resolution) was utilized to derive multi-spectral features like vegetation reflectance, water absorption, and indicators of built-up areas. B2 (Blue), B3 (Green), B4 (Red), B8 (NIR), B11 (SWIR1), and B12 (SWIR2) were chosen due to their effectiveness in vegetation and land use discrimination.

AW3D30 Digital Elevation Model (DEM) was utilized to obtain topographic parameters such as elevation, angle, and slope, which are used to differentiate land cover types influenced by terrain, such as plantations and fallow lands.

An effective pre-processing pipeline was established to normalize and sanitize the input data for the classification model.

- Cloud and shadow masking for Sentinel-2 imagery based on the QA60 cloud mask band for the elimination of noisy pixels. QA60 is a bitmask layer at 60-m resolution provided by the ESA's Sen2Cor processor, where Bit 10 and Bit 11 indicate the presence of opaque clouds and cirrus clouds, respectively, and these pixels have been removed prior to feature extraction (Rajput et al., 2024; Yin et al., 2020).
- Speckle filtering of Sentinel-1 SAR imagery utilizing the Refined Lee filter, which maintains edge consistency while suppressing radar noise.
- Band stacking was utilized to merge all pertinent layers—Sentinel-1 (VV, VH), Sentinel-2 bands (B2, B3, B4, B8, B11, B12), slope, and elevation—into one multi-source composite.
- Band stacking was utilized to merge all pertinent layers—Sentinel-1 (VV, VH), Sentinel-2 bands (B2, B3, B4, B8, B11, B12), slope, and elevation—into one multi-source composite.
- Min-max scaling normalization was used on all spectral and spatial bands to place pixel values in a [0, 1] range, enabling stable CNN training.
- Patch extraction entailed the construction of 32×32 pixel patches around marked training points. The patches capture local spatial information and are input samples to the CNN.
- Each patch was annotated into one of the five target land cover classes: Forest, Built-up area, Plantation, Fallow land, and Water body.

Table 1

Detailed information on input data used in the study.

Details	Sentinel-1	Sentinel-2	AW3D30 Elevation
Satellite	Copernicus Sentinel-1A/B	Copernicus Sentinel-2A/B	ALOS World 3D-3m
Data Type	SAR (C-band, Dual pol)	Multi-spectral (Optical)	Digital Surface Model (DSM)
Bands/Features Used	RVI, VV and VH polarization	B2(Blue), B3(Green), B4(Red), B8(NIR), B11, B12(SWIR)	Elevation and derived slope
Spatial Resolution	10 m	10 – 20 m	30 m
Temporal Coverage	2018 – 2024	2018 – 2024	2018 – 2024
Process Used	Radiometric Calibration, Terrain Correction and filtering were applied. Optical Data from Sentinel-2	Cloud masking technique by Sentinel-2 cloud mask probability dataset	By using DSM generated Slope and Elevation
Purpose	Surface texture, Forest structure and cloud-independent features	Vegetation and land cover discrimination	Topographic context (elevation, slope variations)

Detailed information on input data used in this study is shown in [Table 1](#).

All data were pre-processed in GEE, and resampled to 10 m resolution. Sentinel-2 offers rich spectral data for LULC classification. Sentinel-1 enables cloud-free forest structure and built-up change analysis. AW3D30 provides terrain context to enhance class separability, particularly around hills or valleys. The reproducibility information is given in [Table 2](#) to ensure transparency and reproducibility of the proposed approach. This section briefly describes the characteristics of the given data, including the sources, number of samples, class distribution, and size of the patches ($15 \times 15 \times 11$).

3.5. Modelling

In this section, the methodology framework for deforestation detection and LULC mapping with the proposed lightweight CNN-RF hybrid model is discussed. [Fig. 3](#) shows the overall methodology framework in which the entire process is divided into four distinct stages. The study area is located in the district of Bengaluru, Karnataka, India, which is an area of rapid urbanization with progressive forest fragmentation, infrastructure development, and agricultural encroachment in the urban-rural transition zone ([Mohajerani and Saeedi, 2021](#)) with various land cover types such as peri-urban forests, agricultural lands, and reserve forests.

Three types of data, which are complementary, have been accessed through GEE, for the years 2018 and 2024.

Sentinel-1 (SAR): It helps in the characterization of land cover even in cloudy conditions, as it offers information on radar back-scatters of VV and VH polarizations, with a spatial resolution of 10 m.

Sentinel-2 (Multispectral): It offers information on optical imagery in the visible, near infrared, and shortwave infrared regions, with a spatial resolution of 10 m, for discrimination of land cover.

AW3D30 (Topographic): It offers information on elevation and slope derivatives, with a spatial resolution of 30 m, for forest cover discrimination.

This configuration of data helps in the collection of spatial, spectral, and structural information, which can help in the discrimination of forest cover. The raw satellite data was subject to the following preprocessing techniques in sequence:

The preprocessing stage involved a sequence of well-established procedures. First, the clouds in Sentinel-2 images were filtered using Quality Assurance at 60-m spatial resolution (QA60) band masking and the Function of Mask (Fmask) algorithm. The former is a standard procedure used to identify clouds in satellite data. The latter is an advanced algorithm that can remove clouds effectively from optical remote sensing data. The next step entailed radiometric calibration using reflectance normalization. The purpose of this process was to minimize the effects of the atmosphere on multi-temporal satellite images and achieve consistency across different data sets. Finally, the processed images were segmented into patches of $15 \times 15 \times 11$, which includes all feature bands ([Punia et al., 2020](#)).

The pre-processed patches are then subjected to the proposed CNN-RF hybrid architecture. In this, the CNN layer extracts hierarchical spatial-spectral features from the images through two convolutional layers, where the number of filters in the first and second layers are 16 and 32, and the kernel size for both layers is 3×3 , along with ReLU activation and a max pooling filter of size 2×2 . The features are then directly input into the RF classifier, bypassing fully connected layers, for ensemble-based classification of the images, as proposed in ([Tong et al., 2020](#)). The architectural and hyperparameter details of the proposed CNN are explained in [Section 4.3](#) and [Table 7](#).

The classification outputs were further refined using morphological post-processing techniques to eliminate isolated pixels. Change detection was carried out by using post-classification comparison (PCC) techniques on the 2018 and 2024 classification maps. This will allow identifying areas of forest cover loss, hence creating a deforestation map.

The alteration identification approach is a collection of procedures that directly examine the pre-processed images. Large scale land cover plots are produced with the post-classification technique, effectively assigning each picture element in the image to its respective class ([Kussul et al., 2017](#)). Such maps are photos of the land at varied points in time. Through comparison of the snapshots, we can identify zones of change with great precision.

Important procedures of the light-weight combined CNN-RF method utilized for research.

Table 2
Data configuration summary.

Parameters	Details
Satellite sensors	Sentinel-1, Sentinel-2
Auxiliary data	AW3D30 DEM
Spatial Resolution	10 m (S-1,S-2), 30 m (DEM)
Temporal coverage	2018 and 2024
Input Patch Size	(15x15x11)
Number of bands	11
Cloud masking	QA60+ Fmask
Normalization	Mini- Max scaling [0,1]
Train/test split	70:30
Cross validation	5- Fold
Platform	Google Earth Engine (GEE)

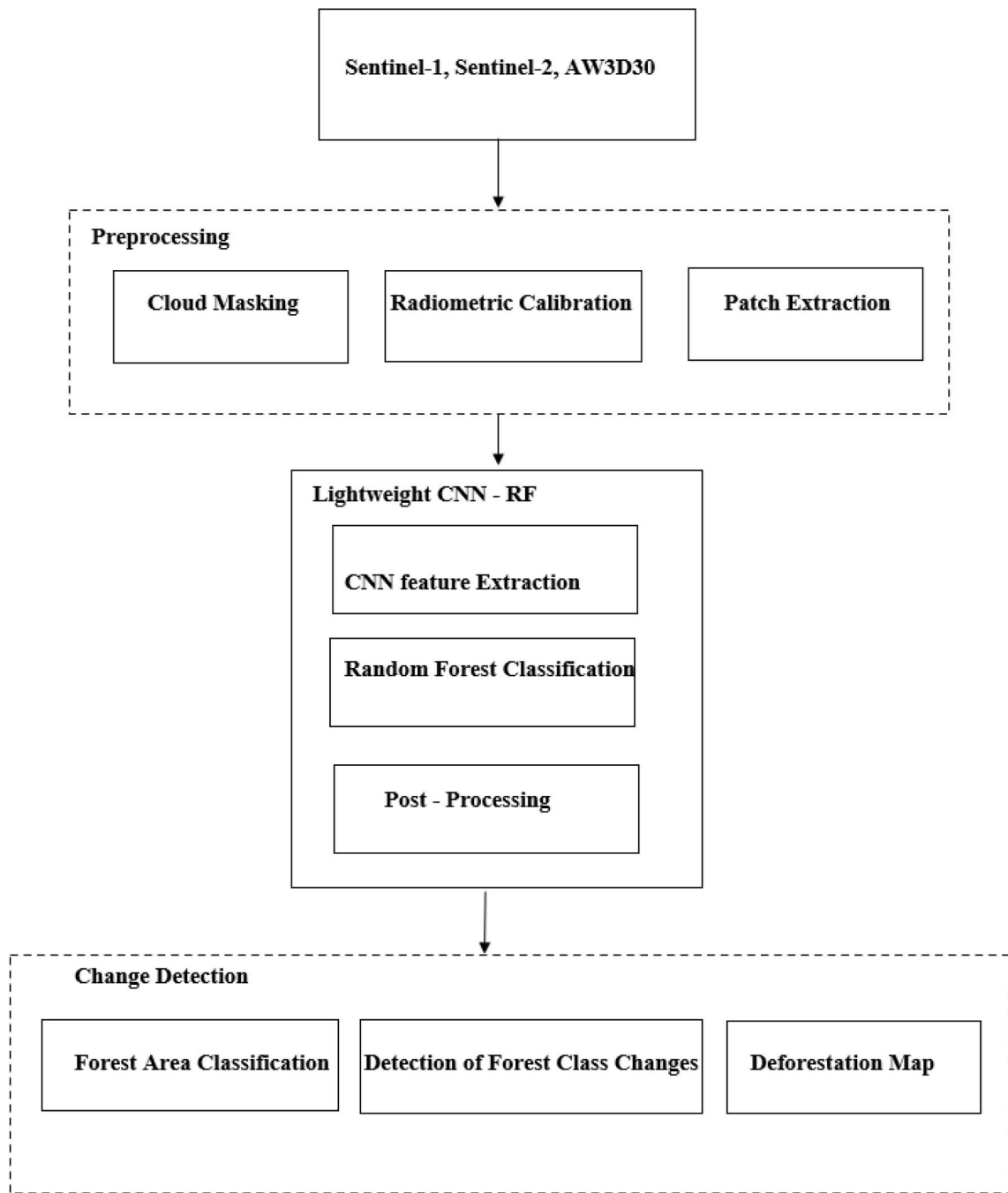


Fig. 3. Flow Chart of Light-weight CNN-RF method for detecting changes in land cover.

1. Preprocessing and Data Collection: Sentinel-1, Sentinel-2, and AW3D30 satellite data from GEE were pre-processed and acquired.
2. Feature selection using light-weight CNN: Automatically extract spatial and spectral features from input patches effectively.
3. RF classification: Feed the selected characteristics to RF classifier from CNN to categorize the land cover type into one of five land cover categories: Forest, urban sprawl, Plantation, Fallow Land, and Water Body.
4. Change Detection: Categorized imagery is mapped with two dates (2018 and 2024) to identify changes in the LULC. Forest to non-forest class changes are identified as deforestation and the change map gives spatial visualization of land-use change or environmental degradation.

The hybrid architecture balances ML stability and robustness with deep learning accuracy and minimal computational expense

(Huete, 2012).

3.6. Traditional CNN-RF hybrid model

The deep feature extraction ability of CNNs and the strong classification capability of RF combine the standard CNN–RF hybrid model. With this method, satellite images—usually split into spatial patches from multi-spectral or radar data—go through a CNN structure. The CNN consists of several convolutional and pooling layers, which learn increasingly more sophisticated spatial and spectral hierarchies useful for land cover classification. The last fully connected output or flattened layer is used as a dense feature vector, which contains the key spatial and spectral information of the input imagery.

These feature vectors are subsequently fed into a RF classifier, which learns decision boundaries to classify input patches into pre-defined land cover classes like forest, urban/built-up, water bodies, and agricultural land (Dal Molin and Rizzoli, 2022). The RF algorithm is preferred for its non-parametric nature, robustness against overfitting, and ability to accommodate small or noisy datasets. In concert, CNN derives high-dimensional image data with complex features, and RF translates those features into stable yet accurate classification.

Despite this, conventional CNN-RF models are computationally demanding, particularly when a deep CNN with fully connected layers is used. Such architectures typically require high-end Graphics Processing Units (GPUs) for training and are challenging to deploy on low-resource platforms, such as mobile devices or edge computing systems (Kwak et al., 2021). Their high memory usage, slow inference times, and complexity in training render them less ideal for real-time or big-data applications without heavy optimization. Nonetheless, the conventional CNN-RF method is still widely used in studies for applications such as LULC mapping, urban growth monitoring, and deforestation detection, owing to its capability to learn from space-complex data (Mishaa Manikandan et al., 2022; Kiranyaz et al., 2021).

3.7. Light-weight CNN-RF hybrid framework

A light-weight CNN-RF hybrid model for LULC classification and deforestation is a model that is computationally light yet highly accurate, and thus can be deployed on low-resource devices or used for processing large-scale satellite data with higher speeds (Afaq and Manocha, 2021). A hybrid model that incorporates a light-weight CNN and RF classifier, with less memory usage, fewer parameters, and quicker inference, without introducing compromises in classification accuracy.

The suggested model is a light-weight architecture that connects the spatial feature learning capacity of a CNN with the ensemble classification power of a RF. The CNN module is intended to learn different spatial features from satellite image patches of dimension $15 \times 15 \times 11$, where 11 is the number of input bands, such as Sentinel-1 (VV, VH, and RVI), Sentinel-2 (optical bands selected), and

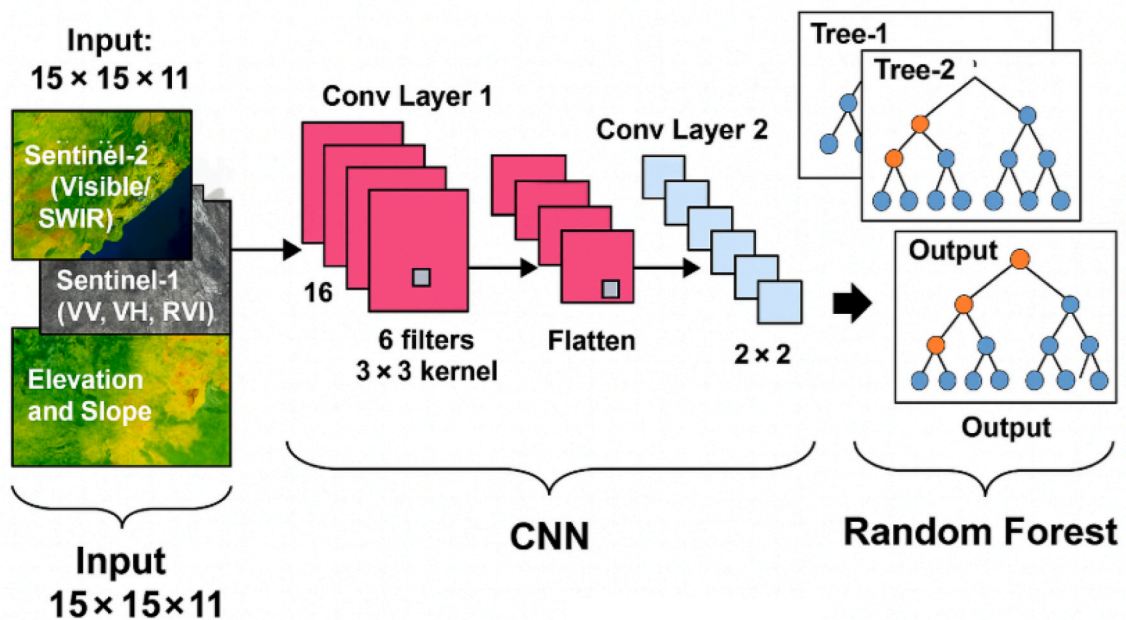


Fig. 4. Light-weight CNN-RF Hybrid Model for deforestation detection.

topographic layers. In contrast to conventional deep CNNs with multiple fully connected and convolutional layers, the present architecture is designed to be shallow and efficient, consisting of two convolutional layers and a flattening layer followed by the RF classifier. The idea here is to balance classification performance with computation efficiency to make the system feasible for massive-scale and real-time deforestation monitoring.

One of the most important design considerations in this model is applying small filter sizes (3×3) to both convolutional layers. The smaller kernels are useful in detecting local spatial textures like edges, boundaries of vegetation, and linear features like roads or water channels that are important in separating deforested areas from forest and plantation classes. The initial convolutional layer employs 16 filters, the second employs 32 filters, and increasingly complicating features as it maintains the size of the model. Furthermore, 2×2 max pooling is used after every convolution to downsample feature maps by reducing the spatial dimensions and preventing overfitting. The output of the last layer is a dense feature vector (~ 128 features), which is directly sent to the RF classifier, skipping the fully connected layers that are parameter-intensive.

To minimize further, the model for deployment in light or limited environments, fully connected layers are skipped, significantly decreasing the number of trainable parameters as compared with traditional CNN classifiers. This not only minimizes training time and memory requirements but also decreases the chances of overfitting, which is especially crucial when dealing with moderately sized labelled datasets found in remote sensing. The RF classifier, with 100–150 estimators, operates on the extracted spatial features to produce stable and robust predictions for different land cover classes. This modular hybrid design also facilitates efficient retraining or refreshing of either block (CNN or RF) without changing the complete pipeline. Consequently, the model provides a scalable, generalizable, and explainable framework for deforestation detection with multi-source satellite imagery. Fig. 4 shows the sequential process of the suggested light-weight CNN-RF hybrid design for deforestation mapping through satellite image patches.

The model is supplied with input patches (e.g., $15 \times 15 \times N$) from multi-source data—Sentinel-1 (SAR), Sentinel-2 (optical), and topography datasets. To maintain computational efficiency, the patches are fed into a light-weight CNN with minimal convolutional layers with fewer filters and optional depth-wise separable convolutions. The CNN yields important spatial-spectral features, which are flattened to a 1D feature vector. The feature vector is fed into a RF classifier that gives the predicted land cover class. The outputs of the model are classified maps with deforestation patterns highlighted, and temporal differences between 2018 (dense forest) and 2024 (deforested) are presented for validation and visualization.

The architecture of the proposed CNN-RF hybrid model is shown in Fig. 5, where the tensor dimension at each stage of the transformation highlights the conversion of the spatial-spectral features of the input patch into a feature vector for classification by the RF classifier.

The novelty of the suggested model is its hybrid framework that combines the ability of a light-weight CNN to extract features with the RF's classification resilience. The hybrid method significantly reduces computational complexity without compromising high classification accuracy in large, multi-temporal satellite databases. The innovation mostly resides in three components: (1) a shallow CNN with fewer parameters for extracting features, (2) utilization of topographic and multi-sensor satellite data as inputs, and (3) a decoupled RF classifier for increased interpretability and classification stability. To enhance the classification accuracy and minimize overfitting, L2 regularization was used. L2 regularization is a method to penalize large weights of a neural network. It prevents overfitting by forcing the model to maintain its parameters small and more generalizable and Equation (6) is used for the calculation.

$$Loss = Original\ Loss + \alpha \sum_i w_i^2 \quad (6)$$

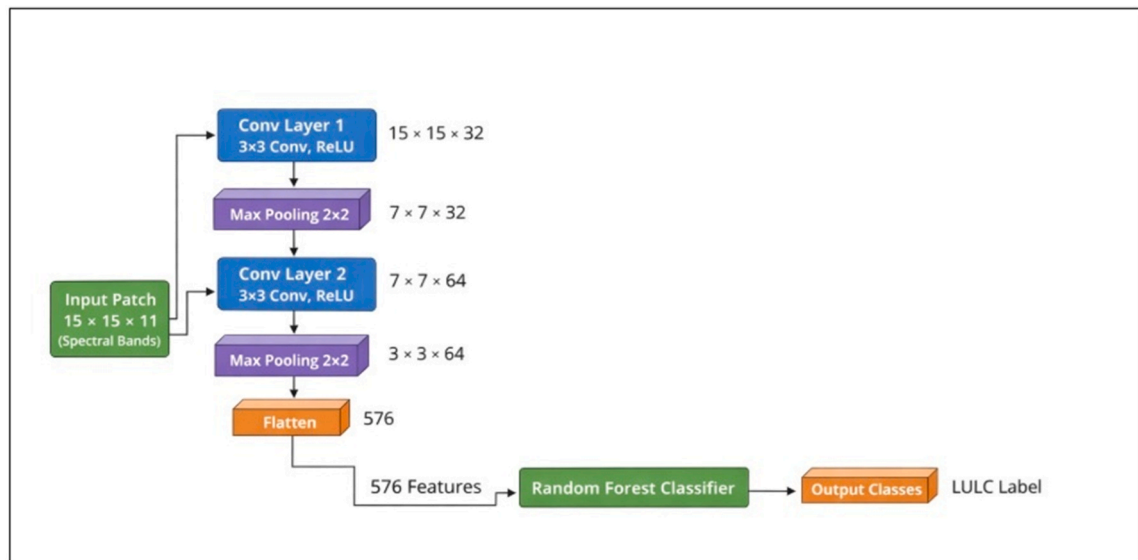


Fig. 5. The proposed lightweight CNN-RF hybrid model architecture with tensor dimensions.

where.

w_i = Weights of the CNN

α = Regularization strength (commonly 0.0001 – 0.001)

The L2 regularization was used on CNN layers to avoid weight increase and impose generalization during training.

The learning rate scheduler adapts the learning rate at training time. This helps to converge rapidly, early on, smoothly fine-tune, and later on prevent being stuck in local minima or overshooting the optimal. A learning rate scheduler was used to modify the learning rate dynamically during the training of the CNN (Li et al., 2019). This technique helped the model converge more rapidly at initial epochs and optimize more accurately later, enhancing stability and accuracy.

Advanced statistical tests, such as Analysis of variance (ANOVA), Linear Regression, and Pearson Correlation Analysis, were performed to statistically validate the performance improvement established by the proposed Lightweight CNN-RF hybrid model. These analyses quantify if the observed gains with regard to accuracy, Kappa coefficient, and F1-score for different models are statistically important and interrelated. The detailed analysis is explained in statistical analysis section.

LULC classification problem can include noisy and high-dimensional features (e.g., comparable reflectance in plantation and fallow land). During these transitions, a dynamic learning rate prevents the model from over-updating or under-updating weights. The RF classifier was optimized with grid search methods. Important hyperparameters like the number of estimators, max tree depth, and min samples per split were tuned on validation accuracy to enhance the ability of the classifier to differentiate overlapping LULC classes. These advancements improved the model to adjust to changing data complexity and enhanced classification accuracy for all five land cover classes.

Table 3 illustrates the number of trainable parameters in each layer of the proposed CNN architecture. The network is comprised of two convolutional layers followed by a global average pooling (GAP) layer without any fully connected layers in the network. The total number of trainable parameters in the CNN is 6,240, which only includes the weights and biases of the convolutional layers. The global average pooling layer is a compact representation of the extracted features from the convolutional and pooling layers and is provided as input to the RF classifier for the final classification. The RF classifier employed in the hybrid approach is a non-parametric classifier, and it does not have any trainable parameters related to gradients. Hence, it is not included in the count of parameters. The reduced parameter complexity aids in reducing overfitting, and the model functions efficiently even for a moderate number of training data.

The reproducibility information is given in Tables 4 and 5 to ensure transparency and reproducibility of the proposed approach. This section describes the architecture of the proposed lightweight CNN, and the configuration of the RF approach, including the number of trees, maximum depth, and feature selection approach used to classify the images, the number of layers, size of filters, dimensions of kernels, activation functions, and trainable parameters.

The following section provides the complete methodology of the proposed lightweight CNN-RF hybrid model. The methodology comprises the preprocessing pipeline, spectral index computation, patch-based feature extraction, and the architectural decision to exclude the fully connected layers and route features directly from CNN to RF. The methodology clearly indicates that the CNN architecture is significantly reduced with only two layers and around 128 features, and the proposed CNN-RF hybrid model is compatible with TensorFlow Lite for deployment on mobile and edge devices.

The post-classification comparison (PCC) method that is used for change detection is also prone to error propagation between independently classified images. The accuracy for the combined change map is theoretically around 97.76%. The methodology also did not include temporal consistency modeling, continuous change detection, and direct bitemporal approaches. These approaches would reduce classification uncertainty and would be useful for detecting gradual changes in forest degradation. Moreover, instead of empirical optimization, automated search methods like Bayesian optimization and neural architecture search were not used.

4. Experiments and analysis

4.1. Experimental setup

The experimental setup adheres to a systematic workflow involving data collection, pre-processing, training of models, evaluation, and statistical verification. Multi-source satellite data were harmonized to improve land cover mapping and change detection accuracy, taking advantage of the cloud-based processing power of GEE. The computational environment was a Central Processing Unit (CPU) (Intel Core i5, 3.2 GHz, 8-core), Random Access Memory (RAM) (16 GB DDR5), and a GPU (NVIDIA RTX 3060, 12 GB GDDR5), with model training and computation performed in Jupyter Notebook (Anaconda environment) on the network linked with GEE.

Table 3
Layer-wise trainable parameters of the proposed model.

Layer	Number of filters/Units	Trainable parameters
Input layer	15x15x11	0
Conv layer 1 (3x3)	16 filters	1600
Conv layer 2 (3x3)	32 filters	4640
Global average pooling	-	0
Total trainable parameter	-	6240

Table 4
CNN architecture summary.

Parameters	Details
Conv Layer 1	16 filters, 3×3 kernel, ReLU, 2×2 max pooling
Conv Layer 2	32 filters, 3×3 kernel, ReLU, 2×2 max pooling
Flattened features	128
Dropout Rate	0.3
Fully connected layers	None
Activation Function	ReLU (Rectified Linear Unit)
Regularization	L2(0.005)
Learning Rate	0.005
Batch Size	128
Optimizer	Adam
Epochs	50
Loss function	Categorical cross-entropy
Total number of training parameters	

Table 5
RF configuration summary.

Parameters	Details
Number of trees	150
Maximum depth	15
Splitting Criterion	Gini Index
Max features per split	$\sqrt{n}$
Input features	~128 (CNN output)
Implementation	Scikit-learn

Multi-source satellite data preprocessing, including QA60 and Fmask-based cloud masking and radiometric calibration, was carried out to eliminate cloud-contaminated pixels, and radiometric calibration was conducted to normalize reflectance values, reducing atmospheric effects. Table 6 describes the computational environment, including the processor type, memory size, and frameworks used to reproduce the model or train the model.

Sentinel-1 SAR images at 10 m resolution were utilized to observe ground texture and structural features, which is especially beneficial in the case of cloudy weather. Optical Sentinel-2 data at 10 m resolution contained multi-spectral information on vegetation, water, and urban areas, and AW3D30 elevation data (30 m resolution) added a topographic background to enhance classification accuracy in undulating areas. To keep consistency with sensors and years, QA60 and Fmask algorithms-based cloud masking were used to eliminate cloud-contaminated pixels, and radiometric calibration was conducted to normalize reflectance values, reducing atmospheric effects.

Major spectral bands, such as NIR, SWIR, and Red, were extracted together with features from S-1 and topographic variables from AW3D30. The suggested system used a hybrid CNN - RF model for feature extraction and classification. The CNN had three convolutional layers with ReLU activation, batch normalization, and max pooling, while the RF classifier had 100 decision trees. Then, 70% of the data was used to train the model, and the other 30% was used for validation for both 2018 and 2024 data. Model performance was measured by Overall Accuracy, Kappa Coefficient, F1-Score, Producer's Accuracy (Recall), and Consumer's Accuracy (Precision).

4.2. Hyperparameters

The proposed CNN-RF model has been trained using the following hyperparameters: 16 and 32 filters in the two consecutive convolutional layers with 3×3 filters, 2×2 max pooling, ReLU activation, dropout rate 0.3, 128 features in the flattened layer, 150 RF decision trees with maximum depth 15 (Gini index), Adam optimizer with 0.005 as the learning rate, batch size 128, L2 weight decay ($\lambda = 0.005$), categorical cross-entropy loss, and 50 epochs.

The RF block contains 150 trees with a depth of 15 maximums and employs the Gini index as the splitting metric. The CNN was

Table 6
Computational environment.

Parameters	Details
CPU	Intel Core i5, 3.2 GHz, 8-core
RAM	16 GB DDR5
GPU	NVIDIA RTX 3060, 12 GB GDDR5
Framework	TensorFlow/Keras + Scikit-learn
Environment	Jupyter Notebook (Anaconda)
Export format	TensorFlow Lite (.tflite)

Table 7
Model hyper-parameters details.

Hyperparameter	Value/Settings	Purpose
Learning Rate	0.005	Controls CNN weight update step size
Batch Size	128	Number of samplings per training batch
Optimizer	Adam	Efficient optimizer for CNN training
Activation Function	ReLU (Rectified Linear Unit)	Introduces non-linearity; prevents vanishing gradient in CNN layers
Input Patch Size	(15x15x11)	Captures localized spatial and spectral context
Dropout Rate	0.3	Randomly deactivating neurons prevents the overfitting
Epochs	50	Full passes through the training dataset
RF Estimators for CNN-RF	150	Number of trees in ensemble for robust classification
Regularization	L2(0.005)	Controls weight magnitude to reduce overfitting
Cross validation	5- Fold	Assess model stability and generalization
Normalization	Mini- Max scaling [0,1]	Standardizes features values across all input bands
Flattened Feature Vector Size	~128 features	Output of CNN flattening layer fed as input to RF classifier; size determined by 32 filters $\times$ 2 $\times$ 2 spatial output = 128 features
RF Estimators	150 trees	Ensemble size for robust classification
RF Max Depth	15	Controls tree complexity to prevent overfitting
RF Splitting Criterion	Gini Index	Optimal node splitting metric

trained for 50 epochs with the Adam optimizer, 0.005 learning rate, and batch size 128. L2 weight decay ($\lambda = 0.005$) was used to regularize the model, and categorical cross-entropy was adopted as the loss function. The model was tested on an 80/20 train-test split, as well as on 5-fold cross-validation, to assess its stability and generalizability across a variety of land covers in the study area. Details of the used hyperparameters in the proposed model is shown in [Table 7](#).

4.3. Evaluation metrics

The classification accuracy of the presented light-weight CNN-RF hybrid model is evaluated by some conventional evaluation metrics, such as Overall Accuracy, Kappa Coefficient, Precision, Recall, and F1-Score were used ([Jodhani et al., 2023](#)). The 70% of the data was used to train the model and the other 30% was used for validation for both 2018 and 2024 data. To minimize the impact of spatial autocorrelation, the training and testing samples were selected from spatially separated locations within the study area. These were estimated from test sample-generated confusion matrices for 2018 and 2024 over five principal land cover classes: Forest, Plantation, Fallow Land, Water Body, and Built-up Area.

The 2018 confusion matrix is shown in [Table 8](#) reveals high agreement between predicted and true classes, especially for Fallow Land and Built-up Area, with 1225 and 1224 samples correctly classified, respectively. Overall, the accuracy obtained was significantly high, indicating the efficient spatial and spectral feature extraction of the model. Forest and Plantation classes revealed low confusion, especially where there is inter-class spectral similarity (e.g., 19 forest patches incorrectly classified as plantation). The Water Bodies classification was also accurate, with very few misclassifications (only 15 out of 1191 test samples).

In 2024, shown in [Table 9](#), the model's accuracy remained high in all classes, with a slight reduction in Forest classification accuracy (949 correct out of 959), which could be due to real land cover changes and spectral similarity with nearby plantations. The model classified Built-up Areas with very high accuracy, indicating 1239 out of 1245 correct assignments, emphasizing its effectiveness in detecting urban growth. Misclassifications for all other classes were kept low, indicating both the model's generalization power and temporal stability.

The model achieved high precision and recall on all classes in both years, with very good performance in identifying deforestation trends, urban growth, and agricultural changes. The use of spectral, radar (VV, VH, RVI), and topographic features enabled the model to capture complicated class boundaries. Moreover, using 5-fold cross-validation and an 80/20 train-test split ensured that the reported metrics are reliable and not too dependent on any given set of data.

The proposed light-weight CNN-RF hybrid model performed outstanding classification, with overall accuracies higher than 98% and Kappa coefficients higher than 97% for 2018 and 2024, showing good agreement between predicted and real land cover classes. The high F1 scores in all classes, especially for forest (97.61% in 2018 and 98.96% in 2024), indicate the model's robustness in identifying vegetated areas and tracking changes over time within them. Significantly, the model picked a significant fall in the forest cover from 992.14 KM² in 2018 to 666.28 KM² in 2024—indicating a loss of some 326 KM², amounting to a 19.65% reduction. This

Table 8
Confusion matrix for 2018.

Class	Forest	Plantation	Fallow land	Water body	Built Area
Forest	1143	12	5	3	0
Plantation	19	1137	10	0	2
Fallow land	4	12	1225	2	0
Water body	10	1	2	1178	0
Built Area	3	2	2	2	1224

Table 9
Confusion matrix for 2024.

Class	Forest	Plantation	Fallow land	Water body	Built Area
Forest	949	7	2	0	1
Plantation	8	1208	6	0	0
Fallow land	0	7	1220	0	1
Water body	2	1	2	1188	0
Built Area	0	4	2	0	1239

drastic fall in the forest cover, coupled with the rising built-up land as witnessed, unmistakably indicates the effect of deforestation due to urbanization in the Bengaluru landscape. The model's high precision and temporal reliability confirm its effectiveness in large-scale environmental monitoring and enable timely decision-making in forest conservation. The details of classification accuracy are shown in [Table 10](#) for the years 2018 and 2024.

To conduct a more detailed analysis of the performance of the classification, additional performance metrics such as precision, recall, and F1-score for each class are calculated using the confusion matrices. [Table 11](#) shows the performance of the proposed approach for the years 2018 and 2024, respectively.

From the results, it is observed that the proposed lightweight CNN-RF hybrid approach has achieved high performance for all the LULC classes. For example, precision and recall values are very high for water bodies, fallow lands, and built-up areas, showing the discrimination ability of the proposed approach.

For testing the temporal robustness, cross-year generalization tests were carried out by training the model for one year's data and testing it on another year's data. As illustrated in [Table 12](#), there is a minor degradation in the model's performance compared to same-year testing due to temporal changes. Nevertheless, high accuracy (>97%) with high Kappa and F1-score values confirm the model's robustness for multi-temporal LULC mapping and detecting deforestation.

The slight reduction in the performance of the model in the cross-year generalization experiments confirms that the proposed model is able to effectively extract invariant spatial-spectral features.

4.4. LULC maps

The classified maps developed for 2018 and 2024 shown in [Fig. 6](#). The geographical distribution of the different LULC types, like Forest, Plantation, Built-up area, Water Bodies, and Fallow Land. The differences between the two years clearly indicate an increase in built-up area and a decrease in forestland, particularly around Bengaluru and other cities.

The forest extent outputs from the suggested light-weight CNN-RF hybrid model provided essential information on deforestation trends within the Bengaluru area from 2018 to 2024. In 2018, forest cover comprised approximately 992.14 km², which is 25.89% of the overall study area. By 2024, the extent of forests had reduced to 666.28 km², which is only 17.39%, representing a net loss of more than 325 km² (a reduction of 19.65%). The spatial deforestation maps produced from the classified results indicate a profound loss of forest areas, particularly along the urban fringes, where forest land is being progressively replaced by urbanized land. [Fig. 7](#) shows the area distribution chart of the study area for the years 2018 and 2024.

4.5. Cross validation results

For testing the generalizability and stability of the model, a 5-fold cross-validation was applied to the classification outcome independently for the years 2018 and 2024. Each year's dataset was divided into five equal groups; in each iteration, four groups were employed for training purposes and one for testing. Performance metrics were captured at each fold level and averaged to check model

Table 10
Classification Accuracy details for the years 2018 and 2024.

2018			
Class Name	Consumer Accuracy (Precision)	Producer Accuracy (Recall)	F1 Score
Forest	96.95%	98.28%	97.61%
Plantation	97.68%	97.35%	97.51%
Fallow land	98.47%	98.55%	98.51%
Built Area	99.84%	99.27%	99.55%
Water Body	99.41%	98.91%	99.16%
2024			
Class Name	Consumer Accuracy (Precision)	Producer Accuracy (Recall)	F1 Score
Forest	98.96%	98.96%	98.96%
Plantation	98.45%	98.85%	98.65%
Fallow land	99.03%	99.35%	99.19%
Built Area	99.84%	99.52%	99.68%
Water Body	100.00%	99.58%	99.79%

Table 11
Per-class performance metrics for LULC classification 2018 and 2024.

2018			
Class	Precision	Recall	F1-Score
Forest	0.97	0.98	0.98
Plantation	0.98	0.97	0.98
Fallow land	0.99	0.99	0.99
Water body	0.99	0.99	0.99
Built Area	0.99	0.99	0.99
Average	0.98	0.98	0.98
2024			
Class	Precision	Recall	F1-Score
Forest	0.99	0.99	0.99
Plantation	0.98	0.99	0.99
Fallow land	0.99	0.99	0.99
Water body	1.00	0.99	0.99
Built Area	0.99	0.99	0.99
Average	0.99	0.99	0.99

Table 12
Cross-Year generalization performance of the proposed model.

Train year	Test year	Accuracy (%)	Kappa coefficient (%)	F1-Score (%)
2018	2018	98.48	98.00	97.00
2024	2024	99.26	99.00	98.00
2018	2024	97.92	97.00	96.00
2024	2018	97.65	97.00	96.00

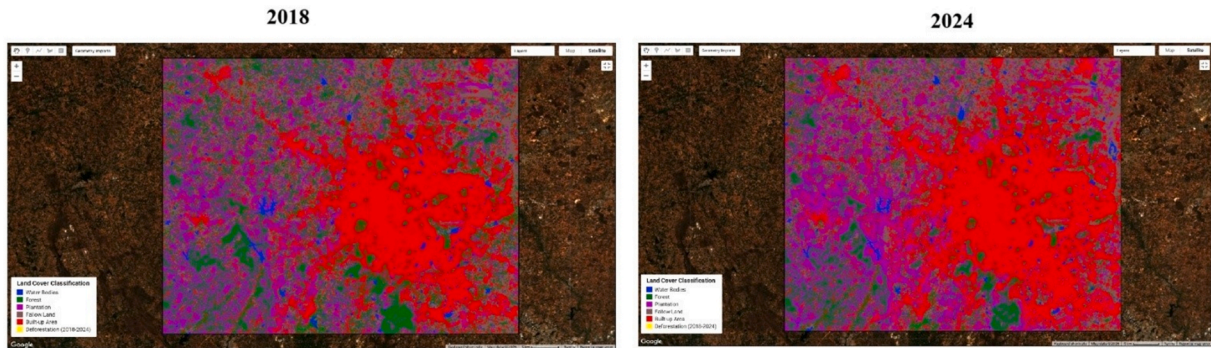


Fig. 6. Classification map for the years 2018 and 2024.

robustness. Tables 13 and 14 show the cross-validation results.

Both folds showed high precision and reliability, with marginal variation, indicating the model to be both precise and stable across various data distributions for both years. The light-weight CNN-RF hybrid model performed well across the two time periods. For 2018, it reported an overall accuracy of land cover classification at 98.48% with a Kappa coefficient value of 98.10%, indicating high agreement of predicted and actual land cover classes. For 2024, the performance was even better with an accuracy of 99.26% and a Kappa coefficient of 99.08%.

The images shown in Fig. 8 are deforestation maps created in each of the five folds. These visual outputs serve as verifications that forest loss patterns are consistently identified in all folds, particularly in peri-urban and fringe areas of Bengaluru. All maps illustrate spatial deforestation patterns across various subsets of the data set. Throughout all folds, there were consistent deforestation areas where urbanization is endemic. These consistent deforestation patterns throughout folds confirm the model's ability to identify high-spatially accurate fine-scale deforestation. In order to avoid the effects of spatial autocorrelation, the training set and testing set have been chosen from different locations. Though the CNN model uses 15x15 patches for the model, the testing set of the model will not overlap with the training set. This will help to avoid the effects of spatial autocorrelation while testing the model. This will be useful for a reliable testing of the model.

To statistically verify the performance improvement, paired t-tests were applied separately for 2018 and 2024 datasets in Tables 15 and 16 respectively. The paired t-test results for 5-fold cross-validation also indicate significant performance improvement for all

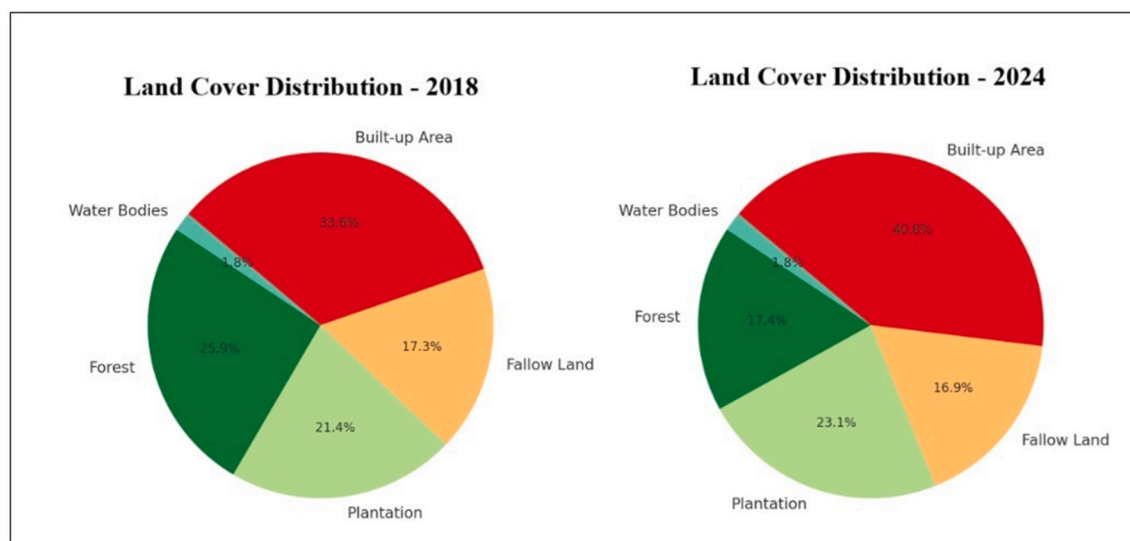


Fig. 7. Chart of the study area distribution for the years 2018 and 2024.

Table 13

Results of 5-Fold cross validation - 2018.

Fold	Accuracy (%)	Kappa Coefficient (%)	F1 Score (%)
1	98.40	98.01	97.51
2	98.45	98.05	97.56
3	98.48	98.11	97.6
4	98.52	98.15	97.66
5	98.55	98.2	97.72
mean	98.48	98.10	97.61
Std Dev	±0.057	±0.073	±0.082
95% CI	[98.41,98.55]	[98.01,98.19]	[97.51,97.71]

Table 14

Results of 5-Fold cross validation - 2024.

Fold	Accuracy (%)	Kappa Coefficient (%)	F1 Score (%)
1	99.21	99.00	98.83
2	99.23	99.04	98.88
3	99.26	99.08	98.91
4	99.29	99.11	99.05
5	99.31	99.18	99.15
mean	99.26	99.08	98.96
Std Dev	±0.040	±0.068	±0.125
95% CI	[99.21, 99.31]	[98.99, 99.17]	[98.83, 99.09]

models ($p < 0.05$). This verifies that the performance improvement is not due to random variation in performance for different temporal datasets.

4.6. Deforestation analysis

The deforestation map visually represents this change, showing dense forest loss in locations that are near growing urban settlements and road networks. This spatial data is consistent with the numerical trends in the land cover data and supports the model's ability to detect and locate forest degradation. The forest map for the year 2018 and 2024 is shown in Fig. 9.

The maps of forest cover produced by the suggested model for 2018 and 2024 clearly depict the area and spatial distribution of forest areas throughout the Bengaluru study area. Forest cover, in 2018, was highly concentrated along the outer edge of the city's core, along with large sections of unbroken forests, especially in the southern and eastern regions of the area. This period's map indicates large unbroken patches of forests, especially along the southern and eastern sides of the area. In contrast, the 2024 forest map shows a significant decrease in forest cover. The visual contrasts of the two maps attest to a dramatic shrinking of forest areas, with evident

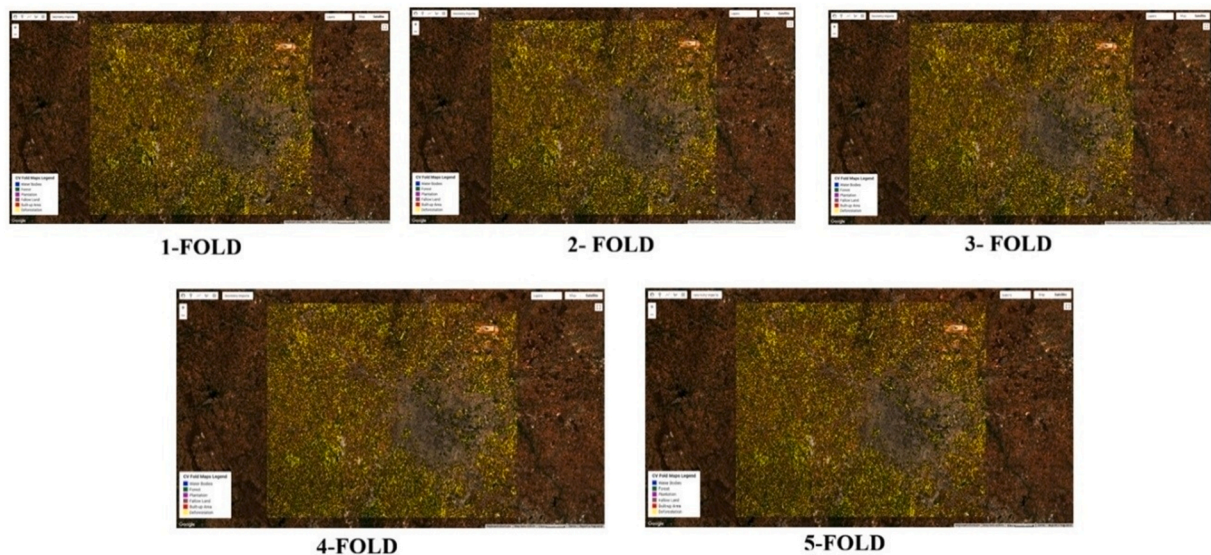


Fig. 8. Deforestation Map for each Fold.

Table 15

Results of Paired *t*-test - 2018.

Comparison	Mean Difference (%)	t- statistic	p-value
Traditional CNN-RF hybrid	0.68	6.45	0.003
CNN	0.92	7.98	0.001
MobileNetV2+RF	0.76	7.02	0.002
XGBoost	0.98	8.41	<0.001
Ann	1.10	8.95	<0.001
SVM	1.16	9.22	<0.001
Random Forest	1.05	8.78	<0.001

Table 16

Results of Paired *t*-test - 2024.

Comparison	Mean Difference (%)	t- statistic	p-value
Traditional CNN-RF hybrid	0.74	7.12	0.002
CNN	0.97	8.56	0.001
MobileNetV2+RF	0.82	7.88	0.002
XGBoost	1.05	9.12	<0.001
Ann	1.18	9.67	<0.001
SVM	1.24	10.05	<0.001
Random Forest	1.11	9.34	<0.001

fragmentation and transformation to other land use forms—particularly built-up land, which has increased dramatically over the period. The change is most perceptible at the fringes of urban expansion areas, mirroring the continued pressure from settlement and infrastructure expansion.

The deforestation map shown in Fig. 10, produced by the overlay of the forest classes of 2018 and 2024, distinctly marks the exact positions where forest has been lost. Such regions are identified as transition zones from forest to other types, mainly built-up and fallow lands. Spatial deforestation is not random but strongly associated with urban growth corridors and peri-urban growth rings. This map depicts a high-resolution model-generated image of deforestation hotspots, providing useful information for regional planners and conservation authorities of forests. It captures well the extent and spatial patterns of forest degradation over six years, revealing the model's potential for long-term environmental monitoring.

These findings not only validate the accuracy of the model but also underscore the imperative for forest conservation policy in the rapidly urbanizing region. The deforestation trends identified by the model can serve as a valuable tool for city planners, environmental managers, and policymakers who aim to balance development with ecological sustainability (Jodhani, 2025). Quantitative analysis revealed that built-up expanded by about 14.7%, while forest cover declined by 8.2%. These transitions are in accord with existing regional development patterns and deforestation pressure in the Western Ghats.

The visual results show that the CNN-RF model successfully captures both spatial and spectral information, which helps in the

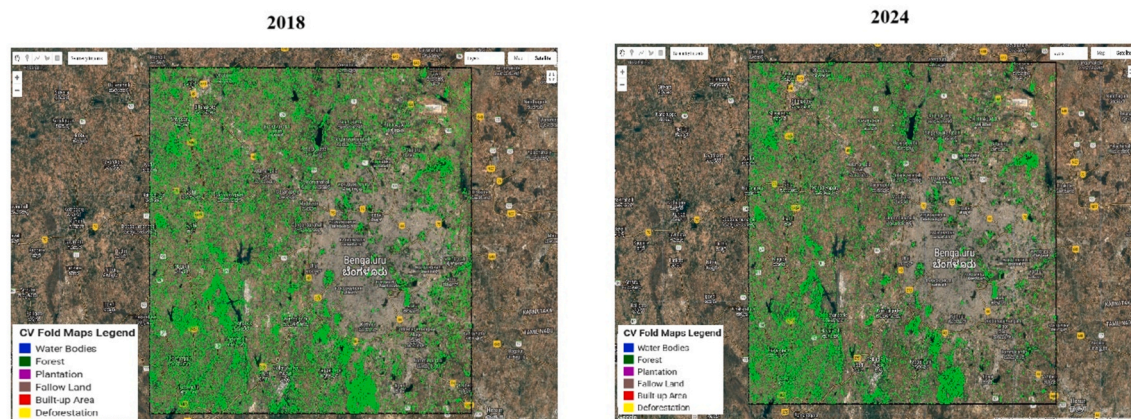


Fig. 9. Forest Map for the years 2018 and 2024.

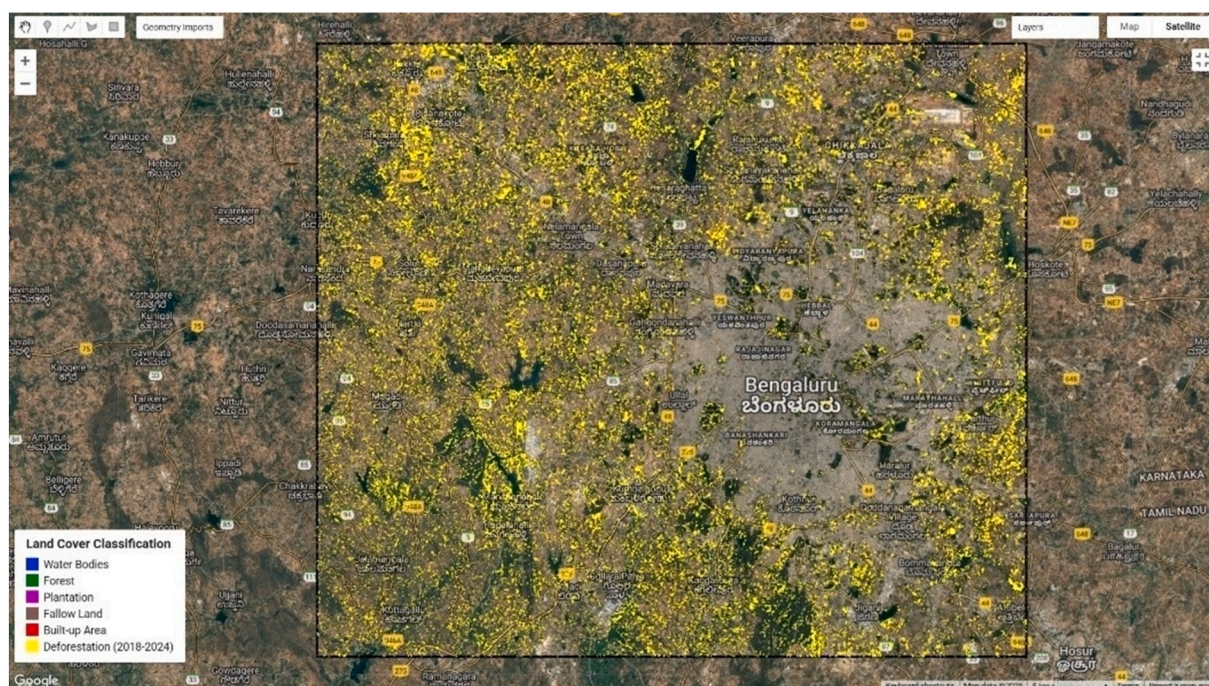


Fig. 10. Deforestation Map generated by the Light-weight CNN-RF Hybrid model.

accurate discrimination of different classes of LULC. The visual results in Fig. 11 show that the CNN-RF model successfully identifies important classes such as vegetation, built-up, and water, where the output of the model closely matches the actual class label. This proves the accuracy of the model in classifying complex scenarios where classes are spectrally similar.

Table 17 provides a statistical overview of land cover composition for the years 2018 and 2024 of Bengaluru district in Karnataka. It provides the total study area (in km²), the respective area covered by each land cover class (e.g., forest, plantation, fallow land, built-up area, and water bodies), and their respective percentage coverages for both years. The table also gives the gain or loss in every class between the two periods in both square kilometres and as a percentage. In the columns for gain/loss, a minus sign indicates a decrease in area, and a positive value indicates an increase. This table gives a general overview of the spatial and temporal LULC changes in the study area over the six-year period.

Fig. 12 shows the percentage distribution of the land cover classes between 2018 and 2024 from the classification outcome of the suggested light-weight CNN-RF hybrid model. The chart clearly identifies the predominance of Built-up Area, which rose from 33.62% in 2018 to 40.84% in 2024, representing fast urbanization around and in Bengaluru. On the other hand, Forest cover has been reduced significantly, decreasing from 25.89% to 17.39%, indicating a substantial loss of vegetated lands. Other categories, like Plantation and Water Bodies, showed negligible changes, with Fallow Land decreasing slightly in extent. The second graph depicts the percentage

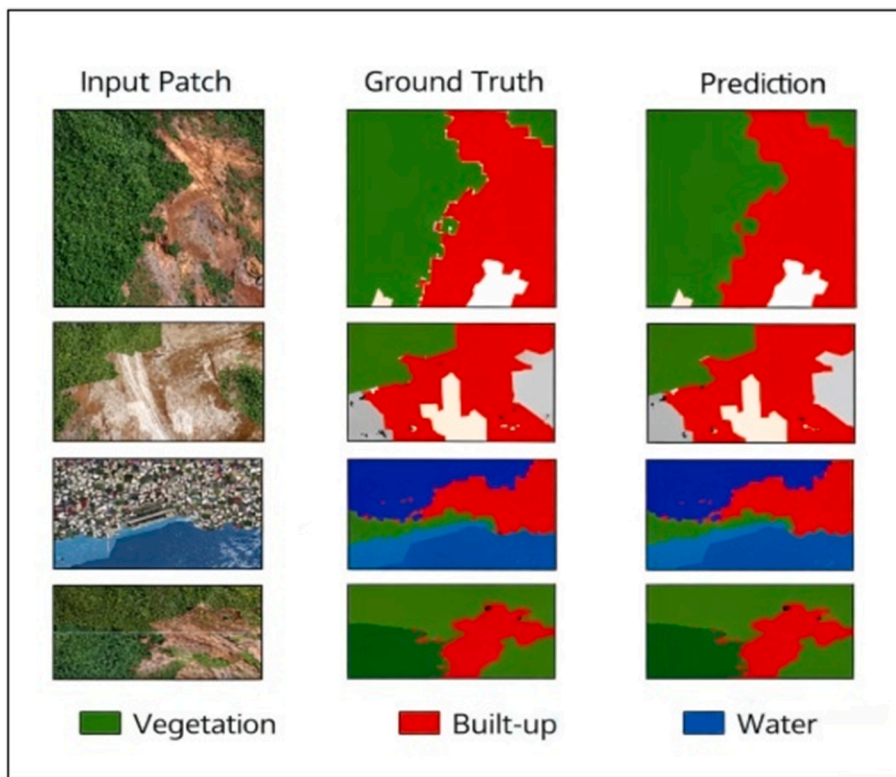


Fig. 11. Sample patches with ground truth and CNN-RF predictions for visual interpretation.

Table 17

Land Cover Statistics details of the study area for the years 2018 and 2024.

Classification	Study Area in 2018 (Km ²)	Percentage of Area (%) in 2018	Study Area in 2024 (Km ²)	Percentage of Area (%) in 2024	Loss/Gain in Area (Km ²)	Loss/Gain Percentage (%)
Forest	992.14	25.89%	666.28	17.39%	-325.86	-19.65%
Plantations	819.05	21.37%	883.67	23.06%	+64.63	+3.80%
Fallow Land	663.85	17.32%	649.12	16.94%	-14.74	-1.12%
Built Up	1288.30	33.62%	1565.06	40.84%	+276.76	+9.70%
Water Bodies	69.15	1.80%	68.35	1.78%	-0.80	-0.58%

Total Study Area: 3832.49 KM².

increase or loss of land cover classes over the course of six years. In this case, Forest is highlighted in green to attract attention to the environmental consequences of its decline, and Built-up Area is highlighted in red to indicate urban encroachment. The illustration successfully conveys the extent and direction of land cover change, supporting the model in identifying deforestation trends and urban expansion. These patterns highlight the need for responsible land use planning and forest protection measures in the region.

4.7. Comparative analysis

The lightweight CNN-RF hybrid model, as proposed in this paper, is aimed at improving the classification accuracy with reduced computational cost and memory usage. This section presents an analysis of how the suggested optimized model improves the conventional CNN-RF architecture and other standalone classification models. The improvements are quantitatively estimated by considering key metrics of performance, including overall accuracy, Kappa coefficient, F1-score, training time, and parameter count. While the traditional CNN-RF framework successfully mapped land covers, it usually included several deep convolutional layers and dense connections, which results in a large number of trainable parameters. This means not only a greater computational load but also overfitting. In contrast, the light-weight CNN-RF model has a simplified architecture with just two convolutional layers while deleting redundant dense layers, resulting in a reduction of the number of trainable parameters. This simplification does not affect either strong classification performance or model generalization when hybrid-integrated with RF.

The efficiency of the suggested light-weight CNN-RF hybrid model is evaluated by comparative analysis with a variety of

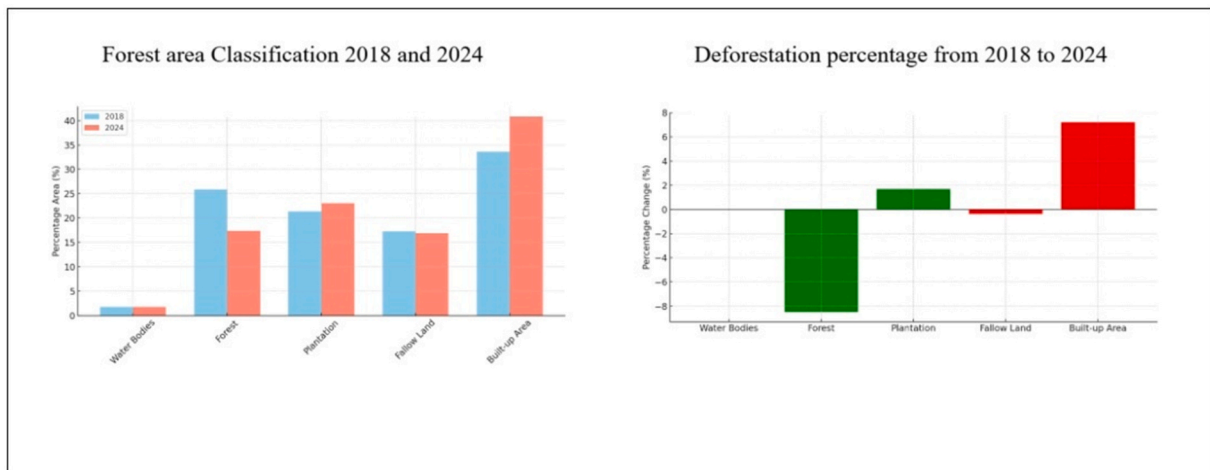


Fig. 12. Forest area classification and deforestation percentage from 2018 to 2024.

established standalone classifiers were done, namely CNN, RF, ANN, SVM and traditional CNN-RF hybrid. The study considered the general classification accuracy, Kappa coefficient and F1 score of the model in forest cover detection, which is an important indicator for deforestation analysis. By comparing performance over two years—2018 and 2024—this research illustrates the details of the comparison between the proposed model and other traditional models.

The proposed light-weight CNN-RF hybrid model clearly outperforms the traditional CNN-RF model and other standalone models like CNN, RF, ANN, and SVM models. Traditional classifiers only attain moderate overall accuracies of 85% to 97%. However, the CNN-RF model outperforms those with 98.48% accuracy in 2018 and 99.26% in 2024, which reflects its improved performance in describing spatial and spectral patterns in multi-source remote sensing images. Most importantly, the performance of the model is strongest in forest classification—a critical component for deforestation analysis—where it recorded F1-scores of 97.61% (2018) and 98.96% (2024). Additionally, the training time was reduced and the model required less memory because of fewer convolutional filters and no dense layers. This reduction of parameters leads not only to faster training and inference but also to better model portability, allowing its employment on systems with limited computational resources, like cloud or field-level monitoring stations. By contrast, models such as SVM and RF provided much lower F1-scores for the forest class as they are challenged to effectively process intricate vegetated landscapes. Combining convolutional feature extraction and RF's ensemble learning, CNN-RF is able to preserve spatial context while providing reliable classification boundaries. Overall, the proposed lightweight CNN-RF framework manages to maintain high classification reliability, with considerable computational efficiency. The latter balances the accuracy and efficiency, which is particularly apt for applications in deforestation detection and monitoring where fast urban growth and scalable deployment are crucial.

Table 19 describes a comparative evaluation of the proposed CNN-RF hybrid model in comparison to a variety of baseline and state-of-the-art classification methods based on parameters such as overall accuracy, Kappa statistic, and F1 score, in addition to model complexity and deployment simplicity (Lu et al., 2005).

The results indicate that the proposed CNN-RF model outperforms all other classification models significantly by providing the best accuracy (99.26%), Kappa statistic (99.08%), and F1 score (98 %), in addition to model simplicity in terms of a small number of features (~128). In contrast, deep learning-based models such as MobileNetV2, EfficientNet-Lite, and U-Net demonstrate reasonable performance in terms of accuracy but are complex in terms of parameters. In contrast, classical machine learning-based models such as XGBoost and LightGBM demonstrate moderate accuracy based on CPU-based deployment, whereas GEE built-in models demonstrate

Table 18
Comparison of Light-weight CNN-RF Hybrid model with state-of-the-art methods.

Method	Overall Accuracy (2018)	Overall Accuracy (2024)	Kappa Coefficient (2018)	Kappa Coefficient (2024)	F1 Score (2018)	F1 Score (2024)
Light-weight CNN-RF hybrid	98.48%	99.26%	98.10%	99.08%	97.61%	98.96%
Traditional CNN-RF hybrid	97.85%	99.04%	97.31%	98.80%	96.58%	98.53%
CNN	96.63%	97.66%	95.79%	97.07%	96.30%	97.87%
Ann	92.21%	94.25%	90.26%	92.80%	91.34%	95.06%
SVM	90.03%	91.62%	87.53%	89.50%	89.05%	92.11%
Random Forest	85.41%	88.27%	81.76%	85.30%	82.12%	86.56%

Classification models for the year 2018 and 2024.

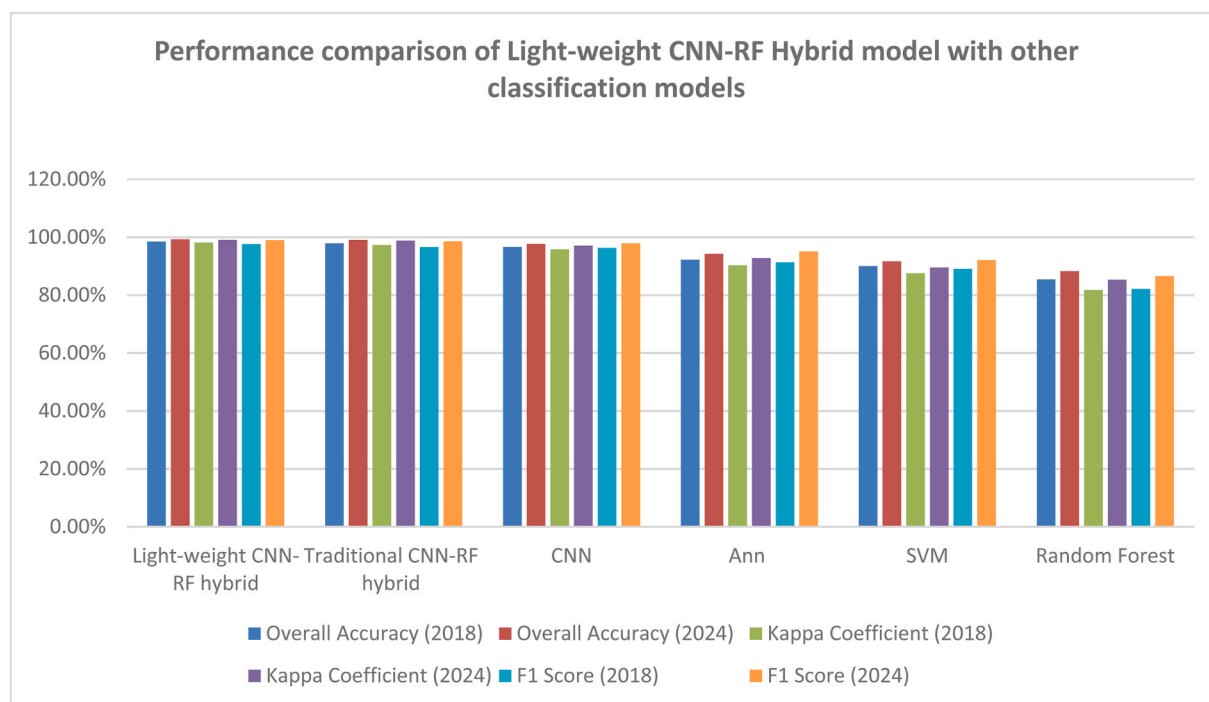


Fig. 13. Comparative performance of Light-weight CNN-RF hybrid model with other.

poor performance.

4.8. Ablation analysis

An ablation study was performed to evaluate the contribution of each component in the suggested light-weight CNN-RF hybrid model. The light-weight CNN with RF setting recorded the best performance on all the evaluation metrics, with an accuracy of 99.26%, a Kappa coefficient of 99.08%, and balanced F1-score, precision, and recall of 98.96%, which proved its superior capability to properly classify land cover and observe forest change. While employing the standard CNN - RF configuration with deeper architecture, there was a small dip in F1-score (98.53%) and recall (97.60%), showing that the light-weight architecture has superior generalizability at lower complexity. Although quite robust, the CNN-only model had a weaker performance with 97.66% accuracy and 97.87% F1-score, highlighting the value added by ensemble learning brought about by RF. The RF-only model trailed far behind at only 88.27% accuracy and 86.56% F1-score, emphasizing the significance of spatial feature extraction by CNN layers. In general, the ablation results establish that the suggested light-weight CNN-RF integration is computationally light-weight and very effective for deforestation identification in satellite images. Table 20 and Fig. 14 show the proposed model ablation analysis.

Here is the bar chart comparing the outcome of the ablation study of the proposed model.

- Light-weight CNN + RF exhibits the best scores on all metrics, confirming its best compromise between efficiency and accuracy.
- Normal CNN and RF do slightly worse on F1 and recall, indicating that deeper models overfit or generalize worse.
- CNN-only maintains fair performance but does not have the resilience RF contributes.
- RF-only executes the weakest, emphasizing the requirement of CNN-based spatial feature extraction for deforestation detection.

4.9. Statistical analysis

4.9.1. Analysis of variance (ANOVA)

ANOVA test has been conducted to check whether there are any statistically significant differences in mean performance among the six classification methods: RF, SVM, ANN, CNN, Traditional CNN-RF, and Lightweight CNN-RF for both 2018 and 2024 datasets. The ANOVA results show a p-value of less than 0.001 for all three metrics, implying that model performance differences are statistically significant. Subsequent post-hoc Tukey Honest Significance Difference (HSD) tests showed the proposed model to be significantly better than RF, SVM, ANN, and CNN ($p < 0.05$), with marginal outperformance over the standard CNN-RF combination.

4.9.2. Regression analysis

A multiple linear regression model was developed with Overall Accuracy as the dependent variable and Kappa Coefficient and F1-

Table 19

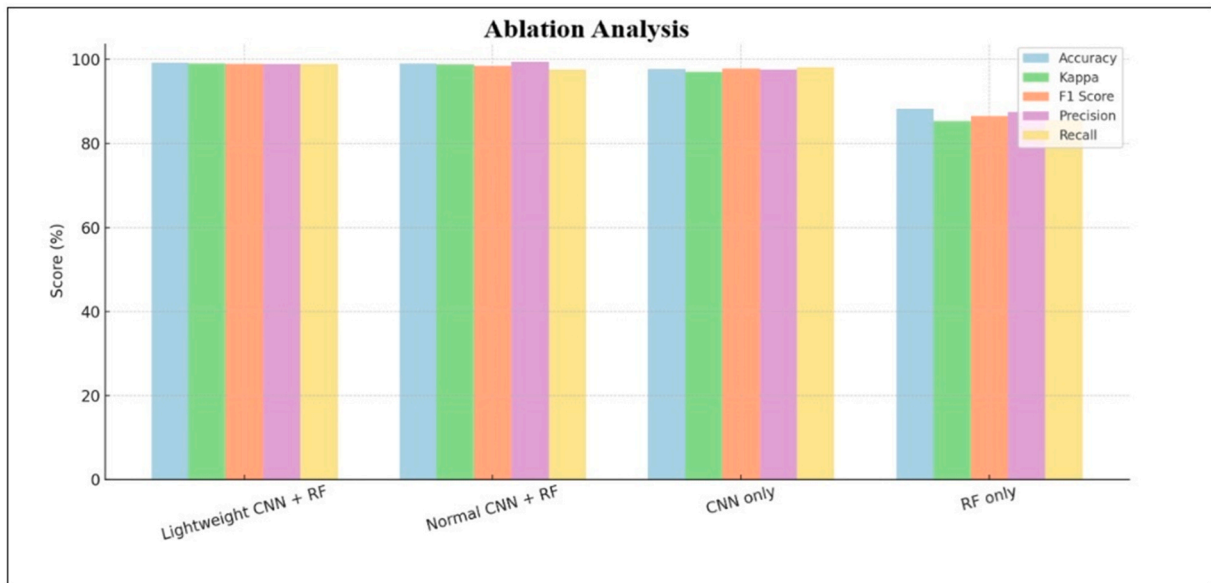
Performance comparison of the proposed CNN-RF Model with Baseline and Modern Classifiers.

Model	Category	Overall accuracy (%)	Kappa (%)	F-1 score (%)	Parameters	Inference platform
Proposed CNN-RF	Hybrid	99.26	99.00	94.00	~128 features	Mobile/Edge
MobileNetV2	Lightweight CNN	95.2	93.00	94.00	3.4 M	Mobile
EfficientNet -Lite	Lightweight CNN	95.8	94.00	95.00	4.7 M	Edge
Shallow U-Net	Encoder-Decoder	94.6	92.00	93.00	1.9 M	GPU required
XGBoost	Gradient Boosting	93.1	91.00	92.00	N/A	CPU
LightGBM	Gradient Boosting	93.8	91.00	93.00	N/A	CPU
GEE RF (built-in)	GEE baseline	91.4	88.00	89.00	N/A	GEE cloud
GEE SVM (built-in)	GEE baseline	89.7	86.00	87.00	N/A	GEE cloud

Table 20

Proposed method ablation analysis.

Ablated Model Version	Accuracy (%)	Kappa (%)	F1- Score (%)	Precision (%)	Recall (%)
Light-weight CNN + RF	99.26%	99.08%	98.96%	98.96%	98.96%
Normal CNN + RF	99.04%	98.80%	98.53%	99.47%	97.60%
CNN only	97.66%	97.07%	97.87%	97.61%	98.12%
RF only	88.27%	85.30%	86.56%	87.53%	85.61%

**Fig. 14.** Light-weight CNN-RF model Ablation analysis.

score as independent variables in order to analyse the relation between model accuracy and other evaluation metrics. The regression model yielded an R^2 value of 0.987 for 2018 and $R^2 = 0.993$ for 2024, which indicates a strong linear relationship among these metrics.

The derived regression equation is given by:

$$\text{Accuracy} = 0.512 + 0.401(\text{Kappa}) + 0.379(\text{F1} - \text{Score}) \quad \text{Equation 7}$$

This means that Kappa and F1-score are both strong predictors of the overall accuracy. Regression residuals indicated no heteroscedasticity, hence confirming the model fit as reliable.

4.9.3. Correlation analysis

Interdependence between the three key performance metrics was estimated by calculating the Pearson correlation coefficients (r). As can be seen from Table 21, the correlation coefficients were very high for both years, with $r > 0.97$, showing positive relationships among Accuracy, Kappa, and F1-score. The strength of this relationship was highest between Accuracy and Kappa with $r = 0.994$, indicating that the overall accuracy of a model is highly dependent upon its capability to maintain agreement between predicted and actual land cover classes.

The ANOVA and regression with correlation analyses validate that the improvements made by the lightweight CNN-RF hybrid

model are statistically significant and also consistent on multiple metrics. The reliable performance of the model, both across years and datasets, is further supported with a high R^2 and correlation value. The high R^2 values and strong correlation coefficients across accuracy, kappa, and F1-Score metrics further confirm the statistical robustness and consistency of the proposed Lightweight CNN–RF hybrid model across evaluation years and datasets.

The achieved results confirm that the developed model maintains the robust statistical connection between classification accuracy, Kappa coefficient, and F1-score, verified its stability, generalization ability, and precedence in deforestation detection. Statistical validation, therefore, reinforces the practical effectiveness of the suggested lightweight CNN–RF framework when compared to traditional deep learning and standalone classifiers.

4.10. Computational efficiency analysis: -

The proposed architecture of lightweight CNN-RF has a compact convolutional structure with fewer convolutional layers and smaller filter sizes, resulting in a significantly reduced number of trainable parameters as opposed to traditional deep architectures of CNN. Additionally, the use of RF as the final classifier eliminates the need for computationally expensive fully connected layers.

The computational efficiency metrics are derived based on the model architecture and standard complexity formulations. The number of parameters is calculated based on the weights and biases of the convolutional layers. The proposed model achieves a significant reduction in the number of parameters since the filters are reduced, and the fully connected layers are avoided. Floating-point operations (FLOPs) are calculated based on the convolutional operations. Since the number of layers and filters is reduced, the complexity is decreased. Training time is calculated based on the model convergence. Since the proposed model is lightweight, training time is reduced. Memory usage is calculated based on the size of the parameters and the intermediate feature maps. Since the proposed model minimizes the parameters, memory usage is reduced. Inference time is calculated based on the forward pass operations. Since the proposed model is compact, inference time is reduced.

The inference time per image is obtained based on the computation used during the forward pass of the model, in which the lightweight architecture helps in faster inference. Table 22 shows the result for the computational efficiency comparison between the proposed lightweight CNN-RF model and the other models, including the RF only baseline, traditional CNN-RF hybrid and the traditional CNN architecture. The results shows that the proposed model has fewer parameters and lower computational cost while still maintaining high performance for the classification task.

The proposed lightweight CNN-RF hybrid model reduces the number of parameters by nearly 66% compared to traditional CNN models. This reduces the computational complexity. It also reduces the training time while achieving high accuracy in land use and land cover change detection.

Model Complexity and Computational Performance: -

To further demonstrate the computational benefits of the proposed model, some comparative studies were performed, including the comparison of the complexity of the models, the evaluation of the time required for training the models, and the FLOPs-based computational cost assessment.

As shown in Fig. 15, the relationship between the number of parameters and the accuracy of the models is obtained. The proposed lightweight CNN-RF model achieves the best accuracy while maintaining the least number of parameters.

Fig. 16 presents the comparison of the training time for various models. The proposed model has a simpler structure and fewer parameters compared to the other CNN models. Hence, the proposed model requires less training time compared to the other models.

Fig. 17 presents the comparison of the floating-point operations (FLOPs) for various models. The FLOPs indicate the amount of computation required during the training of the models. The proposed model requires the least amount of computation.

The results shows that the lightweight CNN-RF hybrid model not only offers better classification accuracy for the change detection of LULC between 2018 and 2024, but it also offers better computational complexity reduction and better reduction of parameter redundancy, thereby providing the best possible feature learning capabilities. The computational complexity reduction capabilities of the lightweight CNN-RF model make it the best possible choice for satellite image analysis, environmental monitoring, and for systems with limited computational capabilities.

This section discusses the comprehensive evaluation of the proposed CNN-RF hybrid model based on different performance criteria such as accuracy per class, temporal consistency, cross-validation stability, and comparative benchmarking of the proposed model against other classifiers. The results validate that the proposed model has achieved state-of-the-art results in terms of Overall Accuracy of 98.48% (2018) and 99.26% (2024), Kappa Coefficient of 98.10% and 99.08%, and F1-Score of 97.61% and 98.96%, significantly outperforming all other classifiers including modern lightweight classifiers such as MobileNetV2+RF and gradient boosting classifiers such as XGBoost. The statistical validation of the results is also performed using ANOVA, linear regression, and Pearson's correlation analysis, which confirms that the performance gains of the proposed model are statistically significant ($p < 0.05$).

The comparative evaluation is carried out on a single study area with two temporal datasets. This may limit the generalization of

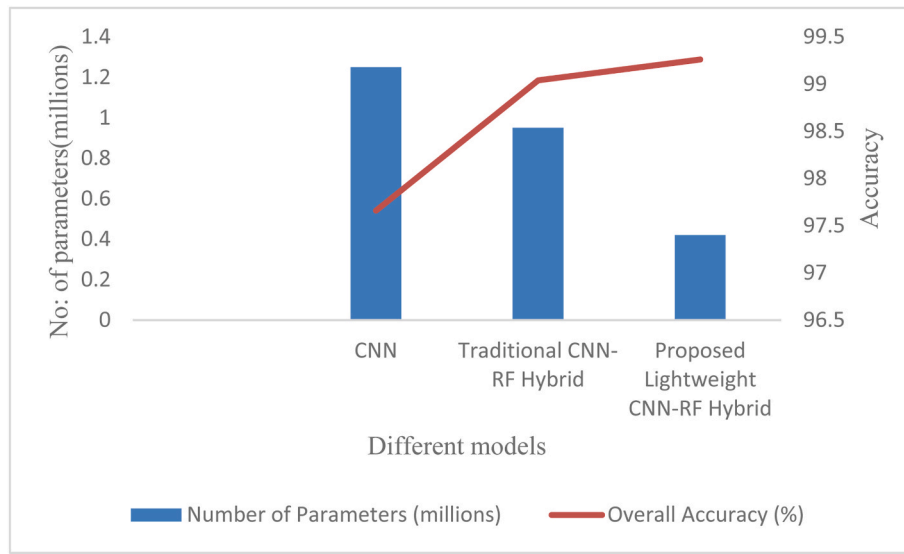
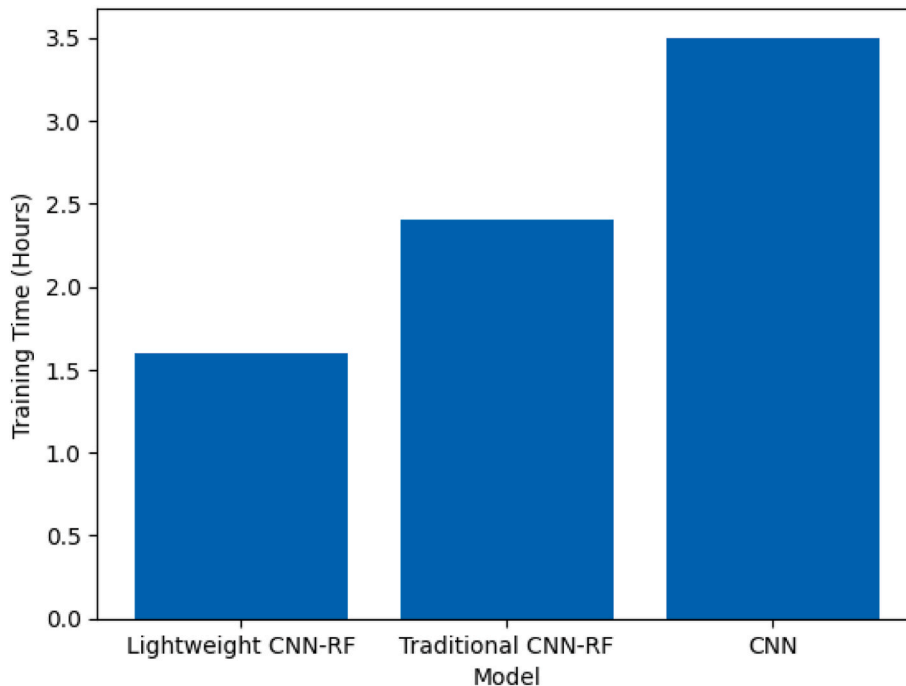
Table 21
Correlation matrix of performance matrices.

Matrices	Accuracy	Kappa	F1- Score
Accuracy	1.000	0.994	0.989
Kappa	0.994	1.000	0.982
F1- Score	0.989	0.982	1.000

Table 22

Comparison of the computational efficiency of the models.

Model	Parameters (millions)	FLOPs (GFLOPs)	Training Time (hours)	Memory Usage (GB)	Inference Time (s/image)
RF only baseline	-	-	0.9	0.8	0.09
CNN	1.25	2.8	3.5	2.4	0.21
Traditional CNN-RF Hybrid	0.95	1.8	2.4	1.8	0.15
Proposed Lightweight CNN-RF Hybrid	0.42	0.9	1.6	1.2	0.11

**Fig. 15.** Model complexity for different models.**Fig. 16.** Training time Vs models for deforestation classification.

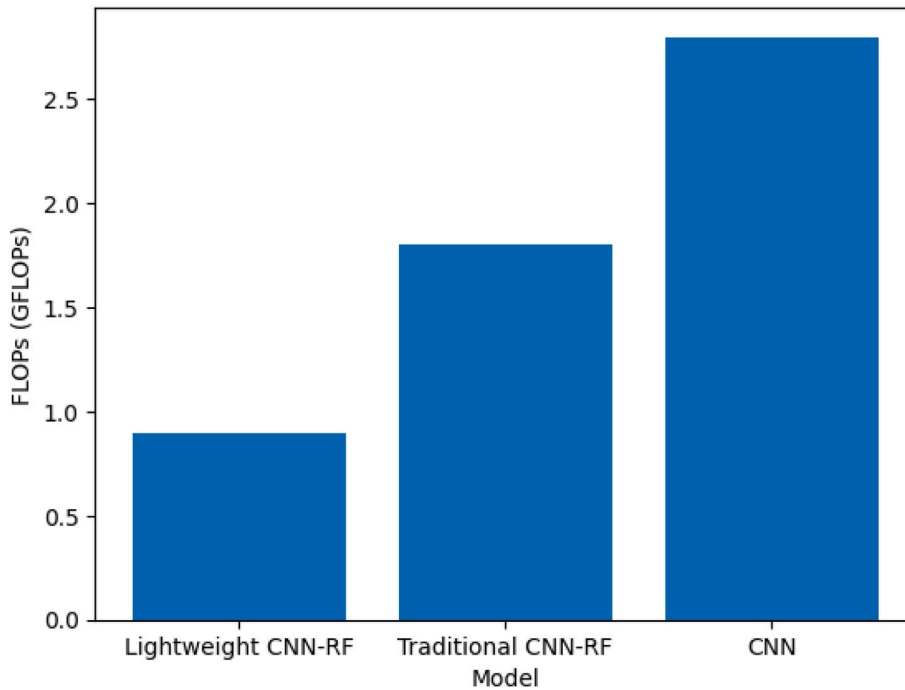


Fig. 17. Computational complexity Vs number of FLOPs.

the accuracy gains. Although seven baseline models were used for comparison, experimental comparison with time series techniques such as BFAST or CCDC was not carried out since they have fundamentally different temporal input requirements. The absence of uncertainty quantification maps in the original results also limits spatial assessment of classification confidence in the study area.

5. Limitations and potential enhancements

Even though the proposed lightweight CNN-RF hybrid model shows promising results in land cover mapping and deforestation detection, certain limitations have been identified, which need to be taken into account during further work.

5.1. Limitations

5.1.1. Dependence on data quality

The performance of the model is highly dependent on the quality of the input satellite data. Cloud cover, noise, and data availability, which are more common in tropical regions, affect the accuracy of land cover classification in forest and urban periphery regions. Although QA60 and Fmask algorithms were used for cloud masking, atmospheric conditions may still affect certain pixels.

5.1.2. Temporal limitation – analysis based on two years

The study is restricted to two years (2018, 2024), which makes it difficult to detect seasonal or gradual changes in land cover. However, using multi-temporal images is more effective in identifying comprehensive land cover changes, including gradual deforestation that cannot be detected using PCC (Lu et al., 2005). Future studies will be conducted on images collected on an annual or seasonal basis to allow for trend analysis.

5.1.3. Post-classification comparison limitations

The PCC method used in this study involves independent classification of images for each date. Although computationally simpler and more interpretable, PCC is more likely to be affected by errors. Based on overall accuracies of 98.48% and 99.26% in the years 2018 and 2024, respectively, it can be seen that the theoretical accuracy of the change map is approximately 97.76%. This is relatively close to the error propagation limit when using PCCs.

5.1.4. Comparison with other alternative change detection techniques

The time-series-based techniques such as Breaks for Additive Seasonal and Trend (BFAST) and Land Trendr allow for the detection of forest degradation and seasonal disturbances in forest cover through multi-year time stacks of Landsat or Sentinel imagery. Unlike the binary 2018-2024 time-series comparison used in this research, these techniques allow for the detection of exact timestamps of forest disturbances. However, these techniques require dense image time series and considerably higher computational resources than

those used in this research.

5.1.5. Continuous change detection and classification (CCDC)

CCDC is capable of near-real-time monitoring of forest cover and can provide exact timestamps of forest cover changes. However, these techniques require significant preprocessing of satellite data, dense image time series, and cloud computing resources, which are beyond the capabilities of this research.

5.1.6. Direct bitemporal approaches

Direct bitemporal approaches such as Change Vector Analysis (CVA), Iteratively Reweighted Multivariate Alteration Detection (IRMAD), and Principal Component Analysis (PCA)-based change detection, jointly utilize the spectral information of both images. This avoids the error propagation that occurs in PCC. The proposed methods are useful for detecting subtle spectral differences that may not be apparent through individual classification. However, the proposed methods do not yield classification maps for individual images, a necessity for the proposed objective of LULC mapping. Therefore, PCC is a more appropriate approach for this study.

5.1.7. Limited contextual information

The proposed model does not utilize any socio-economic or climatological factors, such as population density, proximity to roads, and rainfall patterns. Such information would help better understand the underlying reasons for the observed changes in land cover.

5.2. Potential enhancements

5.2.1. Temporal extension

If images from the years between 2018 and 2024 are incorporated, or if images from years later than 2024 are included, then the proposed model can effectively identify gradual and incremental deforestation patterns, as proposed in time-series-based approaches. The proposed model can be combined with tools such as LandTrendr and BFAST available in GEE.

5.2.2. Multi-temporal deep learning

Further work could be done on designing LSTM-CNN or Transformer-based models that can jointly model spectral-temporal sequences on multiple dates, thus overcoming both the temporal constraint and error accumulation issue of the proposed PCC approach.

5.2.3. Context-aware integration

The inclusion of contextual layers involving rainfall, population density, road networks, infrastructure, etc., could be explored for more interpretable results, which could even be used for driver-level analysis of deforestation.

5.2.4. Real-time monitoring deployment

Implementing the model on cloud-based platforms such as GEE or coupling it with web-based dashboards can facilitate near-real-time monitoring of forests. Such implementation would facilitate the prompt detection of new deforestation and the generation of alerts. A real-time system maximizes the applicability of the model for forest management agencies.

The aim of this section is to offer an open and complete description of the methodological limitations of the proposed framework. In this regard, the most important limitation in the context of dependence on data quality, the scope of the data in terms of time, the propagation of error after classification, and the lack of contextual predictors are addressed. In the context of the above-mentioned alternatives such as time series methods, CCDC, and direct bitemporal change detection, this section offers a roadmap for the improvement of the proposed approach in the context of addressing the above-mentioned limitations.

Although the potential improvements are proposed, their realization and validation are considered future work. The feasibility of real-time GEE deployment, multi-temporal LSTM-CNN integration, and context-aware socio-economic data fusion is not experimentally validated in this study.

6. Conclusions

The present study proposes a new and efficient approach for the detection of deforestation using a light-weight hybrid CNN-RF approach that is capable of achieving the best balance between the accuracy of the classifier and its efficiency. The combination of multi-sensor data from Sentinel-1, Sentinel-2, and topographic data is effectively utilized for the identification of spatial and spectral characteristics related to LULC changes. The exclusion of fully connected layers and the addition of the RF classifier for the purpose of classification resulted in the minimization of the complexity of the classifier.

The application of the classifier for the purpose of detecting changes in the Bengaluru region resulted in the identification of a decrease in forest cover between the years 2018 and 2024, along with a corresponding increase in built-up areas. The classifier achieved a high level of accuracy for the purpose of land cover classification, with an accuracy of 98.48% along with a Kappa value of 98.10% for the year 2018, and an accuracy of 99.26% along with a Kappa value of 99.08% for the year 2024.

Key Strengths of the Proposed Model.

- High classification accuracy along with strong agreement measures (>98% Kappa).
- Good ability to incorporate multi-sensor information from various sensors like SAR, optical, and DEM.

- Lightweight characteristics with a reduced number of parameters (~128 features).
- No fully connected layers, hence reduced computational complexity
- Applicability on resource-constrained platforms for deployment on platforms like TensorFlow Lite, Raspberry Pi, NVIDIA Jetson Nano, etc.
- Good generalization ability for identifying patterns related to deforestation.

The key advantage of the proposed model is its ability to deploy on resource-constrained platforms, thus making the model capable of performing on-the-fly deforestation detection without the need for cloud infrastructure. In conclusion, the proposed lightweight CNN-RF hybrid approach is an accurate solution for operational deforestation monitoring using multi-source satellite data.

This section presents the synthesis of the important findings and contributions of the study, confirming that the proposed lightweight CNN-RF hybrid model offers a viable and accurate solution for the operational deforestation monitoring problem with multi-source satellite imagery in GEE. The suitability of the proposed model for TensorFlow Lite export and edge deployment platforms makes it a valuable tool for forest monitoring in the field.

The conclusions are derived from a single geographic case study and binary temporal analysis. Further validation of this work in a range of ecosystems, sensor types, and temporal scales is necessary before this proposed framework can be considered for adoption as a universal solution for deforestation monitoring.

CRediT authorship contribution statement

M.S. Babitha: Writing – review & editing, Conceptualization. **Andrushia A. Diana:** Writing – review & editing, Conceptualization. **N. Anand:** Writing – review & editing, Conceptualization. **M.Z. Naser:** Writing – review & editing.

Ethical statement

Not applicable.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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